

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\E-30002.pvdb

Date of Analysis : Aug 12, 2026 10:41am

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 520.000 mm ID	
Vessel Design Length, Tangent to Tangent	396.20 cm
Specified Datum Line Distance	5.00 cm
Shell Side Design Temperature	88 °C
Channel Side Design Temperature	66 °C
Shell Side Design Pressure	47.200 kgf/cm ²
Channel Side Design Pressure	36.250 kgf/cm ²
Shop Shell Side Test Pressure	65.438 kgf/cm ²
Field Shell Side Test Pressure	65.438 kgf/cm ²
Shop Channel Side Test Pressure	65.426 kgf/cm ²
Field Channel Side Test Pressure	65.426 kgf/cm ²

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-210 A-1	...	...	K02707	
Tubesheet	SA-516 70	...	...	K02700	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
CH. DISH END	36.287	0.00	59.000	3.1750	No	No
CH. SHELL	36.287	0.00	56.000	3.1750	N/A	No
CH. FLG1	36.250	0.00	52.000	3.1750	N/A	No
SH. FLG1	47.200	0.00	52.000	3.1750	N/A	No
SHELL	47.201	0.00	56.000	3.1750	N/A	No
SH. COVER	47.201	1.05	59.000	3.1750	No	No

Liquid Level: 52.00 cm Dens.: 19.991 kg/m³ Sp. Gr.: 0.020

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	5.0	16.0	14.0	9.9	...	1.00	1.00
Cylinder	50.0	50.0	14.0	14.0	10.1	...	1.00	1.00
Body Flg	61.3	11.0	92.0	80.0	73.2	68.6	1.00	1.00



Body Flg	81.3	11.0	92.0	80.0	75.4	68.6	1.00	1.00
Cylinder	386.1	304.8	14.0	14.0	12.2	...	1.00	1.00
Ellipse	391.2	5.1	16.0	14.0	11.9	4.6	1.00	1.00

Local Stress Analysis Results:

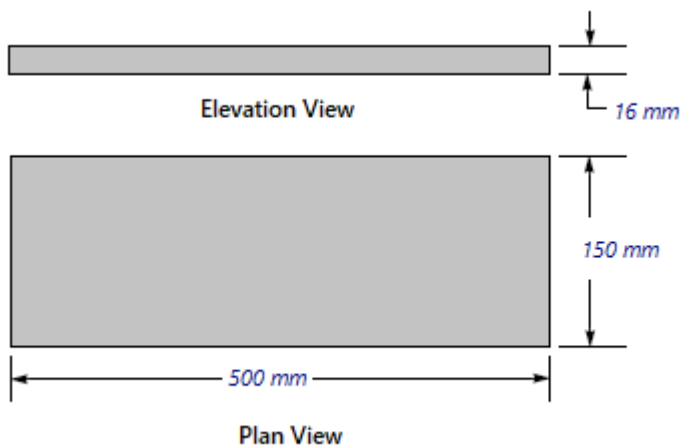
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
TS IN NOZZLE	WRC-297	0.640	Shell	Passed
TS OUT NOZZLE	WRC-297	0.639	Shell	Passed
SH. IN NOZ-1	WRC-297	0.832	Shell	Passed
SH IN NOZ-2	WRC-297	0.832	Shell	Passed
SH OUT NOZ-1	WRC-297	0.832	Shell	Passed
SH OUT NOZ-2	WRC-297	0.832	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	150.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	530.000 mm
Wear Pad Width	225.000 mm
Wear Pad Thickness	10.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	450.000 mm
Baseplate Length	500.000 mm
Baseplate Thickness	16.000 mm
Baseplate Width	150.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	12.000 mm
Web Thickness	12.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	2

Baseplate Sketch



Maximum Tensile Bolt Load 34. kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	3022.18 kgf
Maximum Transverse Saddle Shear Load	252.76 kgf
Maximum Longitudinal Saddle Shear Load	253.35 kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	2877.21 kgf
Maximum Transverse Saddle Shear Load	83.41 kgf
Maximum Longitudinal Saddle Shear Load	24.95 kgf



Weights:

Fabricated - Bare w/o removable internals	2799.5	kg
Shop Test - Fabricated + water (full)	3588.3	kg
Shipping - Fab. + rem. intls.+ shipping app.	2799.5	kg
Erected - Fab. + rem. intls.+ insul. (etc)	2799.5	kg
Empty - Fab. + intls. + details + wghts.	2799.5	kg
Operating - Empty + operating liquid (no ca)	3001.2	kg
Field Test - Empty weight + test medium (full)	3417.1	kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
TS IN NOZZLE	56	...	...	OK	9.39	OK	Passed
TS OUT NOZZLE	56	...	...	OK	9.39	OK	Passed
SH. IN NOZ-1	56	...	...	OK	10.34	OK	Passed
SH IN NOZ-2	56	...	...	OK	10.34	OK	Passed
SH OUT NOZ-1	56	...	...	OK	9.39	OK	Passed
SH OUT NOZ-2	56	...	...	OK	9.39	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 56 Nozzle : SH. IN NOZ-1 [Shellside]
 Minimum MAWP Nozzles : 56 Nozzle : TS OUT NOZZLE [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	TS IN NOZZLE	300.000	-90.000	305.359
20	TS OUT NOZZLE	300.000	90.000	305.359
50	SH. IN NOZ-1	1713.175	90.000	400.050
50	SH IN NOZ-2	3061.175	90.000	400.050
50	SH OUT NOZ-1	1713.175	-90.000	305.359
50	SH OUT NOZ-2	3061.175	-90.000	305.359

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
TS IN NOZZLE	6.000 in	80	WNF	6.625	10.973	318.27	12.00	227.99	600
TS OUT NOZZLE	6.000 in	80	WNF	6.625	10.973	318.27	10.00	227.99	600
SH OUT NOZ-1	6.000 in	80	WNF	6.625	10.973	318.27	12.00	227.99	600
SH OUT NOZ-2	6.000 in	80	WNF	6.625	10.973	318.27	12.00	227.99	600
SH. IN NOZ-1	8.000 in	80	WNF	8.625	12.700	369.08	12.00	238.20	600
SH IN NOZ-2	8.000 in	80	WNF	8.625	12.700	369.08	12.00	238.20	600

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
TS IN NOZZL	SA-106 B	10.973	10.000	10.000	12.000	...
TS OUT NOZZ	SA-106 B	...	10.000	10.000	10.000	...
SH OUT NOZ-	SA-106 B	10.973	10.000	10.000	12.000	...
SH OUT NOZ-	SA-106 B	10.973	10.000	10.000	12.000	...
SH. IN NOZ-	SA-106 B	12.000	10.000	10.000	12.000	...
SH IN NOZ-2	SA-106 B	12.000	10.000	10.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
TS IN NOZZLE	25.000	-90.0	200.00	0.00	CH. SHELL
TS OUT NOZZLE	25.000	90.0	200.00	0.00	CH. SHELL
SH OUT NOZ-1	166.318	-90.0	200.00	0.00	SHELL
SH OUT NOZ-2	301.118	-90.0	200.00	0.00	SHELL
SH. IN NOZ-1	166.318	90.0	200.00	0.00	SHELL
SH IN NOZ-2	301.118	90.0	200.00	0.00	SHELL

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 520.0mm ID X 5.0cm	SA-516 70
1	CYLINDER: 14.0mm THK X 520.0mm ID X 50.0cm	SA-516 70
2	WELD NECK FLANGE: 92.0mm THK X 698.0mm OD	SA-266 2
1	CYLINDER: 14.0mm THK X 520.0mm ID X 304.8cm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 16.0mm THK X 520.0mm ID X 5.1cm	SA-516 70
1	TUBESHEET: 595.0mm X 90.0mm THK	SA-516 70
228	TUBES: 3042.0mm X 19.0mm DIA X 2.1mm THK	SA-210 A-1
2	SADDLE: 150.0mm X 120 DEG	SA-516 70
1	GASKET: 592.0mm OD X 526.0mm ID	...
28	BODY FLANGE BOLTS: 27.0mm DIA	SA-193 B7
56	NUTS FOR BODY FLANGE BOLTS: 27.0mm DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
SH. FLG1	[11]	B	-18	-24	-29	0.891	14.000	1.00	No
SHELL	[8]	B	-18	-24	-29	0.889	14.000	1.00	No
SH. COVER	[10]	B	-18	-26	-29	0.859	14.000	1.00	No
SH. COVER	[7]	B	-14	-28	-29	0.750	16.000	1.00	No
Right Saddle	[24]	B	-29	-35	-29	0.889	10.000	1.00	No
right saddle	[24]	B	-29	-35	-29	0.889	10.000	1.00	No
SH. IN NOZ-1	[1]	B	-23	-32	-29	0.832	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
SH IN NOZ-2	[1]	B	-23	-32	-29	0.832	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
SH OUT NOZ-1	[1]	B	-23	-32	-29	0.832	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
SH OUT NOZ-2	[1]	B	-23	-32	-29	0.832	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Tubesheet: SS	[13]	B	-4	-11	-29	0.868	22.500	1.00	No
Warmest MDMT:			-4	-11					
CH. FLG1	[11]	B	-18	-24	-29	0.891	14.000	1.00	No
CH. DISH END	[10]	B	-18	-26	-29	0.859	14.000	1.00	No
CH. DISH END	[7]	B	-14	-28	-29	0.750	16.000	1.00	No
CH. SHELL	[8]	B	-18	-24	-29	0.889	14.000	1.00	No
TS IN NOZZLE	[1]	B	-23	-43	-29	0.637	12.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
TS OUT NOZZLE	[1]	B	-8	-48	-29	0.364	9.601	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Tubesheet: CS	[14]	B	-4	-11	-29	0.868	22.500	1.00	No
Bolting	[21]		-48						
Warmest MDMT:			-4	-11					
Exchanger Side		Computed MDMT		Required MDMT		Pass/Fail			
Shell		-29.0		-28.9		Pass			
Channel/Tube		-29.0		-28.9		Pass			

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.



UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: CH. FLG1 and SH. FLG1

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	47.2	kgf/cm ²
Channel Side Design Pressure	36.25	kgf/cm ²
Shell Side Design Temperature	87.8	°C
Channel Side Design Temperature	65.6	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM	
Importance Factor for IS-1893	2.000	
Soil Factor	1.000	
Zone Number	2.000	
Percent Seismic for Hydrotest	0.000	
Period of Vibration (User defined) Sec	0.000	
Damping Factor	2.000	



User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/6
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	CH. DISH END
Distance "FROM" to "TO"	5 cm
Inside Diameter	520 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	36.25 kgf/cm ²
Design Temperature Internal Pressure	66 °C
Design External Pressure	0 kgf/cm ²



Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2386	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2507.2	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	52	cm
Liquid Density	709.67	kg/m ³

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	CH. SHELL	
Distance "FROM" to "TO"	50	cm
Inside Diameter	520	mm
Element Thickness	14	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	36.25	kgf/cm ²
Design Temperature Internal Pressure	66	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	52	cm
Liquid Density	709.67	kg/m ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	TS IN NOZZLE	
Dist. from "FROM" Node / Offset dist	25	cm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.864	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	



Detail ID	TS OUT NOZZLE	
Dist. from "FROM" Node / Offset dist	25	cm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	49.975	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	CH. FLG1	
Distance "FROM" to "TO"	11	cm
Flange Inside Diameter	520	mm
Element Thickness	80	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	92	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	36.25	kgf/cm ²
Design Temperature Internal Pressure	66	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	1407.1	kgf/cm ²
Allowable Stress, Operating	1407.1	kgf/cm ²
Allowable Stress, Hydrotest	2294.2	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	2374.8	kgf/cm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	4/6	
Element From Node	40	
Element To Node	50	
Element Type	Flange	
Description	SH. FLG1	
Distance "FROM" to "TO"	11	cm
Flange Inside Diameter	520	mm
Element Thickness	80	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	92	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	47.2	kgf/cm ²
Design Temperature Internal Pressure	88	°C
Design External Pressure	0	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element#	5/6	
Element From Node	50	
Element To Node	60	



Element Type	Cylinder
Description	SHELL
Distance "FROM" to "TO"	304.8 cm
Inside Diameter	520 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	47.2 kgf/cm ²
Design Temperature Internal Pressure	88 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	1407.1 kgf/cm ²
Allowable Stress, Operating	1407.1 kgf/cm ²
Allowable Stress, Hydrotest	2386 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	2461.9 kgf/cm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Saddle
Detail ID	Right Saddle
Dist. from "FROM" Node / Offset dist	45 cm
Width of Saddle	150 mm
Height of Saddle at Bottom	530 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	225 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.5
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm
Element From Node	50
Detail Type	Saddle
Detail ID	right saddle
Dist. from "FROM" Node / Offset dist	259.8 cm
Width of Saddle	150 mm
Height of Saddle at Bottom	530 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	225 mm
Thickness of Wear Plate	10 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.5
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 cm
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	52 cm
Liquid Density	19.991 kg/m ³



Element From Node	50	
Detail Type	Nozzle	
Detail ID	SH. IN NOZ-1	
Dist. from "FROM" Node / Offset dist	85	cm
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	75.406	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	SH IN NOZ-2	
Dist. from "FROM" Node / Offset dist	219.8	cm
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	75.406	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	SH OUT NOZ-1	
Dist. from "FROM" Node / Offset dist	85	cm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.864	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	SH OUT NOZ-2	
Dist. from "FROM" Node / Offset dist	219.8	cm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	600	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	50.864	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	50	
Detail Type	Weight	
Detail ID	Baffle	
Dist. from "FROM" Node / Offset dist	152.4	cm
Miscellaneous Weight	299.99	kgf
Offset from Element Centerline	0	mm

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	SH. COVER	
Distance "FROM" to "TO"	5.08	cm
Inside Diameter	520	mm
Element Thickness	14	mm



Internal Corrosion Allowance	3.175	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	47.2	kgf/cm ²
Design Temperature Internal Pressure	88	°C
Design External Pressure	1.0546	kgf/cm ²
Design Temperature External Pressure	0	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Liquid	
Detail ID	Liquid: 60	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	52	cm
Liquid Density	19.991	kg/m ³

XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
CH. DISH END		5	...	5	...
CH. SHELL		55	...	50	...
CH. FLG1		66.3175	...	11	...
SH. FLG1		86.3175	...	11	...
SHELL		391.118	...	304.8	...
SH. COVER		396.198	...	5.08	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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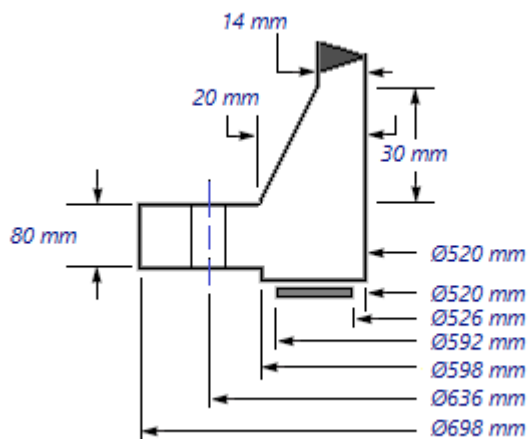


Flange Input Data Values

Description: FLANGE :

CH. FLG1

Flange Type	Integral Weld Neck		
Design Pressure	P	36.25	kgf/cm ²
Design Temperature		66	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	520.000	mm
Flange Outside Diameter	A	698.000	mm
Flange Thickness	t	80.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	30.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	636.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	598.000	mm
Flange Face Inside Diameter	Fid	520.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	592.000	mm
Gasket Inside Diameter	Gi	526.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	526.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	0.0000	
Partition Gasket Design Seating Stress	yPart	0.00	kgf/cm ²





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (36.25 \cdot (520.0/2 + 3.175)) / (1407.13 \cdot 1.0 - 0.6 \cdot 36.25) + Ca$$

$$= 10.0612 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 10.825) / (263.175 + 0.6 \cdot 10.825)$$

$$= 56.485 \text{ kgf/cm}^2$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 14.0) / (260.0 + 0.6 \cdot 14.0)$$

$$= 73.397 \text{ kgf/cm}^2$$

Corroded Flange ID,	Bcor = B+2*ci	526.350 mm
Corroded Large Hub,	g1Cor = g1-ci	16.825 mm
Corroded Small Hub,	g0Cor = go-ci	10.825 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	38.000 mm

Gasket Contact Width,	N = (Go - Gi)/2	32.825 mm
Basic Gasket Width,	bo = N/2	16.413 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	10.209 mm
Gasket Reaction Diameter,	G = Go - 2 * b	571.582 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 571.582^2 \cdot 36.25$$

$$= 93015.430 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P$$

$$= 2 \cdot 10.2088 \cdot \pi \cdot 571.582 \cdot 3.75 \cdot 36.25$$

$$= 49839.270 \text{ kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 526.35^2 \cdot 36.25$$

$$= 78876.305 \text{ kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 93015 - 78876$$

$$= 14139.129 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(93015 + 49839 + 0, 0)$$

$$= 142854.703 \text{ kgf}$$

$$= 186006.656 \text{ kgf, Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 534.33 \cdot 10.2088 \cdot \pi \cdot 571.582 + 0.0 \cdot 0.0 \cdot 526.0$$

$$= 97952.281 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((186007 + 0 + 4 \cdot 0 / 571.58) / 1754, 97952 / 1754)$$

$$= 106.058 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 27.0 + 6 \cdot 80.0 / (3.75 + 0.5)$$

$$= 166.941 \text{ mm}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 636.0 \cdot \sin(\pi / 28)$$

$$= 71.209 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{71.209 / (2 \cdot 27.0 + 80.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	106.058	115.879	
Radial Distance between Hub and Bolts:	38.000	38.000	
Radial Distance between Bolts and Edge:	30.500	31.000	
Circ. Spacing between the Bolts:	65.000	71.209	166.941

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 115.879 \cdot 1753.82 / (534.33 \cdot \pi(592.0 + 526.0))$$

$$= 10.829 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1753.82 \cdot 115.8786$$

$$= 203229.78 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 186007 - 93015$$

$$= 92991.22 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (636.0 - 571.5824) / 2$$

$$= 32.2088 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((636.0 - 526.35) / 2 + 32.2088)$$

$$= 43.5169 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (636.0 - 526.35 - 16.825) / 2$$

$$= 46.4125 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	78876.	46.4125	1.0000	3661.
Face Pressure, Mt	14139.	43.5169	1.0000	615.
Gasket Load, Mg	92991.	32.2088	1.0000	2995.
Gasket Seating, Matm	203230.	32.2088	1.0000	6546.

Total Moment for Operation, Mo 7271. kgf-m.
 Total Moment for Gasket seating, Mg 6546. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 75.483 mm
 Hub Ratio, h/h0 = HL / H0 0.397
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.554

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.861



Factor V per 4.16.5 0.328
Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 698.0/526.35 \\ &= 1.326 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:

$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.326^2 \cdot \log_{10}(1.326))) / (1.326^2 - 1)) / (1.326 - 1) \\ &= 7.031 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.326^2(1 + 8.55246 \cdot \log_{10}(1.326)) - 1) / ((1.0472 + 1.9448 \cdot 1.326^2) \cdot (1.326 - 1)) \\ &= 1.786 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.326^2(1 + 8.55246 \cdot \log_{10}(1.326)) - 1) / (1.36136(1.326^2 - 1) \cdot (1.326 - 1)) \\ &= 7.727 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.326^2 + 1) / (1.326^2 - 1) \\ &= 3.637 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 7.727 \cdot 10.825^2 \cdot 75.483 / 0.328 \\ &= 208548.047 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.861/75.483 \\ &= 0.011 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (80.0 \cdot 0.011 + 1) / 1.786 + 80.0^3/208548.047 \\ &= 3.526 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.0 \cdot 7271/526.35) / (3.5257 \cdot 16.825^2) \\ &= 1384.16 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.0 \cdot 6546/526.35) / (3.5257 \cdot 16.825^2) \\ &= 1246.05 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.2164 \cdot 7271/526.35) / (3.5257 \cdot 80.0^2) \\ &= 135.69 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.2164 \cdot 6546/526.35) / (3.5257 \cdot 80.0^2) \\ &= 122.16 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (7.0314 \cdot 7271 / (80.0^2 \cdot 526.35)) - 3.6365 \cdot 136 \\ &= 1024.29 \text{ kgf/cm}^2 \end{aligned}$$



Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (7.0314 \cdot 6546 / (80.0^2 \cdot 526.35)) - 3.6365 \cdot 122$$

$$= 922.09 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1384 + \max(136, 1024)) / 2$$

$$= 1204.22 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (1246 + \max(122, 922)) / 2$$

$$= 1084.07 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 186007 / 115.8786$$

$$= 1605.19 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (97952 / 115.8786)$$

$$= 845.30 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	1384.16	2110.70	1246.05	2110.70
Radial Flange	135.69	1407.13	122.16	1407.13
Tangential Flange	1024.29	1407.13	922.09	1407.13
Maximum Average	1204.22	1407.13	1084.07	1407.13
Bolting	1605.19	1753.82	845.30	1753.82

Minimum Required Flange Thickness 73.152 mm

Estimated M.A.W.P. (Operating) 52 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 112.2 kg
 Estimated Unfinished Weight of Forging at given Thk 145.2 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6545.8 / 1.0 \cdot 100000.023 \cdot 0.328 / (2.902 \cdot 2067026 \cdot 10.825^2 \cdot 75.483 \cdot 0.3)$$

$$= 0.702 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 7271.3 / 1.0 \cdot 100000.023 \cdot 0.328 / (2.902 \cdot 2041068 \cdot 10.825^2 \cdot 75.483 \cdot 0.3)$$

$$= 0.790 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 6545.8 / 1.0 \cdot 100000.023 \cdot 0.328 / (3.526 \cdot 2067026 \cdot 10.825^2 \cdot 75.483 \cdot 0.3)$$

$$= 0.578 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 7271.3 / 1.0 \cdot 100000.023 \cdot 0.328 / (3.526 \cdot 2041068 \cdot 10.825^2 \cdot 75.483 \cdot 0.3)$$

$$= 0.650 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.891, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C



Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1399.18	2110.70	1236.05	2110.70
Radial Flange	183.13	1407.13	161.78	1407.13
Tangential Flange	864.46	1407.13	763.67	1407.13
Maximum Average	1131.82	1407.13	999.86	1407.13
Bolting	1605.19	1753.82	847.47	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 68.986 mm

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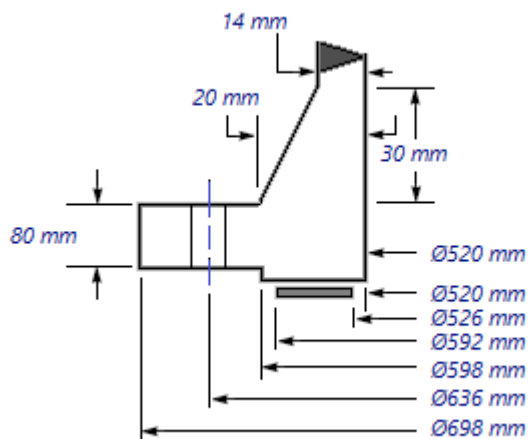


Flange Input Data Values

Description: FLANGE :

SH. FLG1

Flange Type	Integral Weld Neck		
Design Pressure	P	47.20	kgf/cm ²
Design Temperature		88	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	520.000	mm
Flange Outside Diameter	A	698.000	mm
Flange Thickness	t	80.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	20.0000	mm
Length of Hub	h	30.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	1407.13	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1407.13	kgf/cm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	636.000	mm
Nominal Bolt Diameter	a	27.0000	mm
Type of Threads		TEMA Metric	
Number of Bolts		28	
Flange Face Outside Diameter	Fod	598.000	mm
Flange Face Inside Diameter	Fid	520.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	592.000	mm
Gasket Inside Diameter	Gi	526.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	526.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	0.0000	
Partition Gasket Design Seating Stress	yPart	0.00	kgf/cm ²





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Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\ &= (47.2 \cdot (520.0/2 + 3.175)) / (1407.13 \cdot 1.0 - 0.6 \cdot 47.2) + Ca \\ &= 12.1841 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 10.825) / (263.175 + 0.6 \cdot 10.825) \\ &= 56.485 \text{ kgf/cm}^2 \end{aligned}$$

Hub Small End Hub MAPnc:

$$\begin{aligned} &= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 14.0) / (260.0 + 0.6 \cdot 14.0) \\ &= 73.397 \text{ kgf/cm}^2 \end{aligned}$$

Corroded Flange ID,	Bcor = B+2*ci	526.350 mm
Corroded Large Hub,	g1Cor = g1-ci	16.825 mm
Corroded Small Hub,	g0Cor = go-ci	10.825 mm
Code R Dimension,	R = ((C-Bcor)/2)-g1cor	38.000 mm

Gasket Contact Width,	N = (Go - Gi)/2	32.825 mm
Basic Gasket Width,	bo = N/2	16.413 mm
Effective Gasket Width,	b = Cb*sqrt(bo)	10.209 mm
Gasket Reaction Diameter,	G = Go - 2 * b	571.582 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 571.5824^2 \cdot 47.2 \\ &= 121112.492 \text{ kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P \\ &= 2 \cdot 10.2088 \cdot \pi \cdot 571.5824 \cdot 3.75 \cdot 47.2 \\ &= 64894.160 \text{ kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot Bcor^2 \cdot P \\ &= \pi/4 \cdot 526.35^2 \cdot 47.2 \\ &= 102702.383 \text{ kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - Hd \\ &= 121112 - 102702 \\ &= 18410.109 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(Wo,1 + Wo,2 + H'p, 0) \\ &= \max(121112 + 64894 + 0, 0) \\ &= 186006.656 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp \\ &= 534.33 \cdot 10.2088 \cdot \pi \cdot 571.582 + 0.0 \cdot 0.0 \cdot 526.0 \\ &= 97952.281 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa) \\ &= \max((186007 + 0 + 4 \cdot 0 / 571.58) / 1754, 97952 / 1754) \\ &= 106.058 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t / (m + 0.5) \\ &= 2 \cdot 27.0 + 6 \cdot 80.0 / (3.75 + 0.5) \\ &= 166.941 \text{ mm} \end{aligned}$$

Actual Circumferential Bolt Spacing [Bs]:



$$= C \cdot \sin(\pi / n)$$

$$= 636.0 \cdot \sin(\pi / 28)$$

$$= 71.209 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{71.209 / (2 \cdot 27.0 + 80.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	106.058	115.879	
Radial Distance between Hub and Bolts:	38.000	38.000	
Radial Distance between Bolts and Edge:	30.500	31.000	
Circ. Spacing between the Bolts:	65.000	71.209	166.941

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 115.879 \cdot 1753.82 / (534.33 \cdot \pi(592.0 + 526.0))$$

$$= 10.829 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1753.82 \cdot 115.8786$$

$$= 203229.78 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 186007 - 121112$$

$$= 64894.16 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (636.0 - 571.5824) / 2$$

$$= 32.2088 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((636.0 - 526.35) / 2 + 32.2088)$$

$$= 43.5169 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (636.0 - 526.35 - 16.825) / 2$$

$$= 46.4125 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	102702.	46.4125	1.0000	4767.
Face Pressure, Mt	18410.	43.5169	1.0000	801.
Gasket Load, Mg	64894.	32.2088	1.0000	2090.
Gasket Seating, Matm	203230.	32.2088	1.0000	6546.

Total Moment for Operation, Mo 7658. kgf-m.
 Total Moment for Gasket seating, Mg 6546. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 75.483 mm
 Hub Ratio, h/h0 = HL / H0 0.397
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.554

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.861
 Factor V per 4.16.5 0.328



Factor f per 4.16.5

1.000

Stress factor from Table 4.16.4 [K]:

$$\begin{aligned} &= A / B \\ &= 698.0/526.35 \\ &= 1.326 \end{aligned}$$

Stress factor from Table 4.16.4 [Y]:

$$\begin{aligned} &= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1) \\ &= (0.66845 + (5.7169(1.326^2 \cdot \log_{10}(1.326))) / (1.326^2 - 1)) / (1.326 - 1) \\ &= 7.031 \end{aligned}$$

Stress factor from Table 4.16.4 [T]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1)) \\ &= (1.326^2(1 + 8.55246 \cdot \log_{10}(1.326)) - 1) / ((1.0472 + 1.9448 \cdot 1.326^2) \cdot (1.326 - 1)) \\ &= 1.786 \end{aligned}$$

Stress factor from Table 4.16.4 [U]:

$$\begin{aligned} &= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1)) \\ &= (1.326^2(1 + 8.55246 \cdot \log_{10}(1.326)) - 1) / (1.36136(1.326^2 - 1) \cdot (1.326 - 1)) \\ &= 7.727 \end{aligned}$$

Stress factor from Table 4.16.4 [Z]:

$$\begin{aligned} &= (K^2 + 1) / (K^2 - 1) \\ &= (1.326^2 + 1) / (1.326^2 - 1) \\ &= 3.637 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$\begin{aligned} &= U \cdot g_0^2 \cdot h_0 / V \\ &= 7.727 \cdot 10.825^2 \cdot 75.483 / 0.328 \\ &= 208548.047 \text{ mm}^3 \end{aligned}$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.861/75.483 \\ &= 0.011 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (80.0 \cdot 0.011 + 1) / 1.786 + 80.0^3/208548.047 \\ &= 3.526 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g_1^2) \\ &= (1.0 \cdot 7658/526.35) / (3.5257 \cdot 16.825^2) \\ &= 1457.77 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g_1^2) \\ &= (1.0 \cdot 6546/526.35) / (3.5257 \cdot 16.825^2) \\ &= 1246.05 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.2164 \cdot 7658/526.35) / (3.5257 \cdot 80.0^2) \\ &= 142.91 \text{ kgf/cm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.2164 \cdot 6546/526.35) / (3.5257 \cdot 80.0^2) \\ &= 122.16 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (7.0314 \cdot 7658 / (80.0^2 \cdot 526.35)) - 3.6365 \cdot 143 \\ &= 1078.77 \text{ kgf/cm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:



$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (7.0314 \cdot 6546 / (80.0^2 \cdot 526.35)) - 3.6365 \cdot 122$$

$$= 922.09 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (S_{Ho} + \max(S_{Ro}, S_{To})) / 2$$

$$= (1458 + \max(143, 1079)) / 2$$

$$= 1268.27 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (S_{Ha} + \max(S_{Ra}, S_{Ta})) / 2$$

$$= (1246 + \max(122, 922)) / 2$$

$$= 1084.07 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 186007 / 115.8786$$

$$= 1605.19 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (97952 / 115.8786)$$

$$= 845.30 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	1457.77	2110.70	1246.05 2110.70
Radial Flange	142.91	1407.13	122.16 1407.13
Tangential Flange	1078.77	1407.13	922.09 1407.13
Maximum Average	1268.27	1407.13	1084.07 1407.13
Bolting	1605.19	1753.82	845.30 1753.82

Minimum Required Flange Thickness 75.413 mm

Estimated M.A.W.P. (Operating) 52 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 112.2 kg
 Estimated Unfinished Weight of Forging at given Thk 145.2 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 6545.8 / 1.0 \cdot 100000.023 \cdot 0.328 / (3.097 \cdot 2067026 \cdot 10.825^2 \cdot 75.483 \cdot 0.3)$$

$$= 0.658 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 7658.0 / 1.0 \cdot 100000.023 \cdot 0.328 / (3.097 \cdot 2028090 \cdot 10.825^2 \cdot 75.483 \cdot 0.3)$$

$$= 0.785 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot M_a / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{amb} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 6545.8 / 1.0 \cdot 100000.023 \cdot 0.328 / (3.526 \cdot 2067026 \cdot 10.825^2 \cdot 75.483 \cdot 0.3)$$

$$= 0.578 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot M_o / B_{sc} \cdot C_{nvfac} \cdot V / (\lambda \cdot E_{op} \cdot g_o^2 \cdot h_o \cdot K_i)$$

$$= 52.14 \cdot 7658.0 / 1.0 \cdot 100000.023 \cdot 0.328 / (3.526 \cdot 2028090 \cdot 10.825^2 \cdot 75.483 \cdot 0.3)$$

$$= 0.689 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.891, Temperature Reduction per Fig. UCS 66.1 = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -24 °C



Min Metal Temp. w/o impact per UG-20(f)

-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1480.98	2110.70	1236.05	2110.70
Radial Flange	193.84	1407.13	161.78	1407.13
Tangential Flange	914.99	1407.13	763.67	1407.13
Maximum Average	1197.99	1407.13	999.86	1407.13
Bolting	1606.43	1753.82	847.47	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 71.857 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
CH. DISH END		36.287	16	3.175	520	1407.1	1
CH. SHELL		36.287	14	3.175	520	1407.1	2
CH. FLG1		36.25	92	3.175	520	1407.1	3
SH. FLG1		47.2	92	3.175	520	1407.1	4
SHELL		47.201	14	3.175	520	1407.1	5
SH. COVER		47.201	16	3.175	520	1407.1	6

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
CH. DISH END		36.25	58.5259	75.363	14	9.87179	1
CH. SHELL		36.25	56.4476	73.3975	14	10.0684	2
CH. FLG1		36.25	51.5704	51.5307	80	73.152	3
SH. FLG1		47.2	51.5704	51.5307	80	75.4126	4
SHELL		47.2	56.4837	73.3975	14	12.1843	5
SH. COVER		47.2	58.562	75.363	14	11.8927	6

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

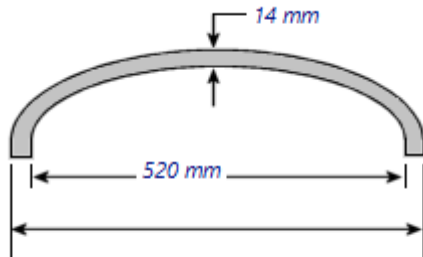
Shell Side MAWP = 50 kgf/cm²
 Shell Side MAPnc = 52 kgf/cm²
 Channel Side MAWP = 50 kgf/cm²
 Channel Side MAPnc = 52 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: CH. DISH END

Elliptical Head Design Information:							
Design Pressure		P	36.287 kgf/cm²	Int. Design Temperature		66 °C	
Material			SA-516 70	Ext. Design Temperature		0 °C	
Length			5.000 cm	External Pressure Chart		CS-2	
				UNS Number		K02700	
Int. Corr. All.		c	3.1750 mm	Allowable Stress (ope)		S	1407.1 kgf/cm²
Ext. Corr. All.		ce	0.0000 mm	Allowable Stress (amb)		Sa	1407.1 kgf/cm²
Inside Diameter		Di	520.000 mm	Actual Stress		871.9 kgf/cm²	
Spec. Min. Thk t _e , or sa			14.0000 mm	Spec. Nominal Thk.		16.000 mm	
Surface Area			4170.0 cm²	SG of Contents		0.710	
Element Volume			29024.2 cm³	Weight of Contents		20.6 kgf	
Empty Weight			53.1 kgf	Operating Weight		73.7 kgf	
				Comp. All. Ope. Stress		1459.1 kgf/cm²	
				Aspect Ratio		2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.250 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (36.287 \cdot 526.35 \cdot 0.984) / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 36.287)$$

$$= 6.6968 + 3.1750 = 9.8718 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.037 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 10.825) / (0.984 \cdot 526.35 + 0.2 \cdot 10.825)$$

$$= 59 - 0 = 59 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 14.0) / (1.0 \cdot 520.0 + 0.2 \cdot 14.0)$$

$$= 75 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (36.287 \cdot (0.984 \cdot 526.35 + 0.2 \cdot 10.825)) / (2 \cdot 1.0 \cdot 10.825)$$

$$= 871.895 \text{ kgf/cm}^2$$



Straight Flange Required Thickness:

$$\begin{aligned} &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)} \\ &= (36.287 \cdot 263.175) / (1407.13 \cdot 1.0 - 0.6 \cdot 36.287) + 3.175 \\ &= 10.068 \text{ mm} \end{aligned}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.037 kgf/cm²](#)

$$\begin{aligned} &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)} \\ &= (1407.13 \cdot 1.0 \cdot 12.825) / (263.175 + 0.6 \cdot 12.825) \\ &= 67 - 0 = 67 \text{ kgf/cm}^2 \end{aligned}$$

Factor K, corroded condition [Kcor]:

$$\begin{aligned} &= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6 \\ &= (2 + (526.35 / (2 \cdot 133.175))^2) / 6 \\ &= 0.984201 \end{aligned}$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 12.448 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 14.0$, $t_r = 9.297$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.859$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 16.0$, $t_r = 9.619$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	50.328 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.750	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

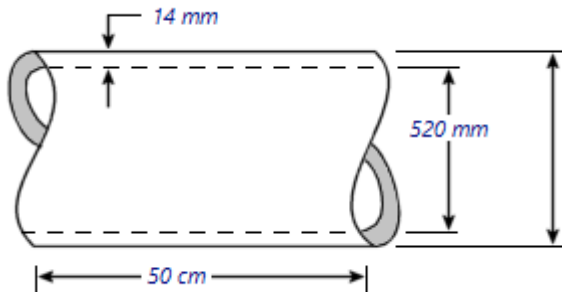
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.500 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.250 mm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: CH. SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	36.287 kgf/cm²	Int. Design Temperature	66 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		50.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	520.000 mm	Actual Stress	904.0 kgf/cm²	
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.	14.000 mm	
Surface Area		8608.0 cm²	SG of Contents	0.710	
Element Volume		106186.1 cm³	Weight of Contents	75.4 kgf	
Empty Weight		91.0 kgf	Operating Weight	166.4 kgf	
			Comp. All. Ope. Stress	1459.1 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (36.287 \cdot 263.175) / (1407.13 - 0.6 \cdot 36.287)$$

$$= 6.8934 + 3.1750 = 10.0684 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.037 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 - 0.6 \cdot 36.287) \cdot 10.0684 / (263.175 + 0.6 \cdot 10.0684)$$

$$= 56 - 0 = 56 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 - 0.6 \cdot 36.287) \cdot 10.0684 / (260.0 + 0.6 \cdot 10.0684)$$

$$= 73 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (36.287 \cdot (263.175 + 0.6 \cdot 10.0684)) / (1.0 \cdot 10.0684)$$

$$= 903.977 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 2.622 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 9.619, c = 3.175 mm, E* = 1.0



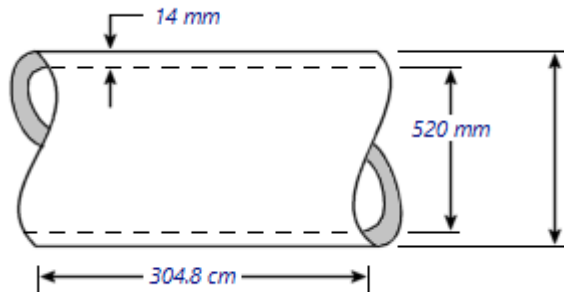
Thickness Ratio = $tr * E^* / (tg - c) = 0.889$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	50.328 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.889	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-24 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	47.201 kgf/cm²	Int. Design Temperature	88 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		304.800 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm²
Inside Diameter	Di	520.000 mm	Actual Stress	1175.9 kgf/cm²	
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.	14.000 mm	
Surface Area		52474.1 cm²	SG of Contents	0.020	
Element Volume		436065.5 cm³	Weight of Contents	8.7 kgf	
Empty Weight		554.8 kgf	Operating Weight	563.5 kgf	
			Comp. All. Ope. Stress	1459.1 kgf/cm²	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (47.201 \cdot 263.175) / (1407.13 - 0.6 \cdot 47.201)$$

$$= 9.0093 + 3.1750 = 12.1843 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 - 0.6 \cdot 10.825) / (263.175 + 0.6 \cdot 10.825)$$

$$= 56 - 0 = 56 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 - 0.6 \cdot 14.0) / (260.0 + 0.6 \cdot 14.0)$$

$$= 73 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (47.201 \cdot (263.175 + 0.6 \cdot 10.825)) / (1.0 \cdot 10.825)$$

$$= 1175.862 \text{ kgf/cm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 2.622 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 14.0, t_r = 9.621, c = 3.175 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.889$, Temp. Reduction = 6 °C

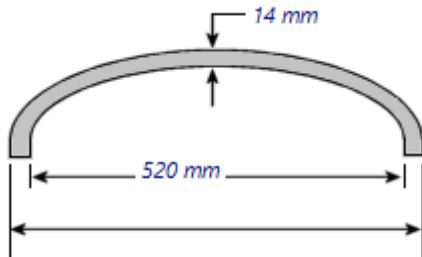
Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	50.337 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.889	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-24 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: SH. COVER

Elliptical Head Design Information:					
Design Pressure	P	47.201 kgf/cm ²	Int. Design Temperature		88 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		5.080 cm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.1750 mm	Allowable Stress (ope)	S	1407.1 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1407.1 kgf/cm ²
Inside Diameter	Di	520.000 mm	Actual Stress		1134.1 kgf/cm ²
Spec. Min. Thk t _e , or sa		14.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		4183.9 cm ²	SG of Contents		0.020
Element Volume		29194.1 cm ³	Weight of Contents		0.6 kgf
Empty Weight		53.2 kgf	Operating Weight		53.8 kgf
			Comp. All. Ope. Stress		1459.1 kgf/cm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.250 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (47.201 \cdot 526.35 \cdot 0.984) / (2 \cdot 1407.13 \cdot 1.0 - 0.2 \cdot 47.201)$$

$$= 8.7177 + 3.1750 = 11.8927 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.001 kgf/cm²

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 10.825) / (0.984 \cdot 526.35 + 0.2 \cdot 10.825)$$

$$= 59 - 0 = 59 \text{ kgf/cm}^2$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 1407.13 \cdot 1.0 \cdot 14.0) / (1.0 \cdot 520.0 + 0.2 \cdot 14.0)$$

$$= 75 \text{ kgf/cm}^2$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (47.201 \cdot (0.984 \cdot 526.35 + 0.2 \cdot 10.825)) / (2 \cdot 1.0 \cdot 10.825)$$

$$= 1134.131 \text{ kgf/cm}^2$$



Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (47.201 \cdot 263.175) / (1407.13 \cdot 1.0 - 0.6 \cdot 47.201) + 3.175$$

$$= 12.184 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.001 kgf/cm²

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (1407.13 \cdot 1.0 \cdot 12.825) / (263.175 + 0.6 \cdot 12.825)$$

$$= 67 - 0 = 67 \text{ kgf/cm}^2$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (526.35 / (2 \cdot 133.175))^2) / 6$$

$$= 0.984201$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 12.448 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 14.0$, $t_r = 9.299$, $c = 3.175$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.859$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -26 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 16.0$, $t_r = 9.621$, $c = 3.175$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.75$, Temp. Reduction = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	50.337 kgf/cm ²
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.750	Temperature Difference	14 °C
UCS-66(c) red.	No	Computed Min. Temp.	-28 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 6.500 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.250 mm

EXTERNAL PRESSURE RESULTS for: SH. COVER

Elliptical Head From 60 to 70 Ext. Chart: CS-2 at 0 °C

Elastic Modulus from Chart: CS-2 at 0 °C: 2038903 kgf/cm²

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
10.825	548.00	50.62	0.0027436	1124.48

$$MAEP = B / (K_0 \cdot D_o / t) = 1124.4806 / (0.9 \cdot 50.6236) = 24.6807 \text{ kgf/cm}^2$$



Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.419	548.00	386.22	0.0003596	366.61

$$MAEP = B / (K0 * Do / t) = 366.6099 / (0.9 * 386.2157) = 1.0547 \text{ kgf/cm}^2$$

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	65.438 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	61.360 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	66.938 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	55.371 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	67.496 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	67.496 kgf/cm ²
App 27-4	= M.A.W.P.	50.337 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	65.426 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	47.125 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	66.938 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	55.361 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	51.837 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	51.837 kgf/cm ²
App 27-4	= M.A.W.P.	50.328 kgf/cm ²

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
CH. DISH END	1573.3	2386.0	0.659	65.48
CH. SHELL	1631.2	2386.0	0.684	65.48
SHELL	1631.5	2386.0	0.684	65.49
SH. COVER	1573.6	2386.0	0.659	65.49

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS IN NOZZLE	Nozzle	1203.20	1203.20	1.000
TS IN NOZZLE	Pad	1407.13	1407.13	1.000
TS OUT NOZZLE	Nozzle	1203.20	1203.20	1.000
TS OUT NOZZLE	Pad	1407.13	1407.13	1.000
SH. IN NOZ-1	Nozzle	1203.20	1203.20	1.000
SH. IN NOZ-1	Pad	1407.13	1407.13	1.000
SH IN NOZ-2	Nozzle	1203.20	1203.20	1.000
SH IN NOZ-2	Pad	1407.13	1407.13	1.000
SH OUT NOZ-1	Nozzle	1203.20	1203.20	1.000
SH OUT NOZ-1	Pad	1407.13	1407.13	1.000
SH OUT NOZ-2	Nozzle	1203.20	1203.20	1.000
SH OUT NOZ-2	Pad	1407.13	1407.13	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
CH. DISH END	1407.13	1407.13	1.000
CH. SHELL	1407.13	1407.13	1.000
CH. FLG1	1407.13	1407.13	1.000
SH. FLG1	1407.13	1407.13	1.000
SHELL	1407.13	1407.13	1.000
SH. COVER	1407.13	1407.13	1.000

Minimum

1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1203.20	1203.20	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
TS IN NOZZLE	680.31	2202.47	0.309
TS OUT NOZZLE	680.31	2202.47	0.309
SH. IN NOZ-1	726.94	2202.47	0.330
SH IN NOZ-2	726.94	2202.47	0.330
SH OUT NOZ-1	680.43	2202.47	0.309
SH OUT NOZ-2	680.43	2202.47	0.309

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	65.438 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	61.360 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	66.938 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	55.371 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	67.496 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	67.496 kgf/cm ²
App 27-4	= M.A.W.P.	50.337 kgf/cm ²

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	65.426 kgf/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	47.125 kgf/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	66.938 kgf/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	55.361 kgf/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	51.837 kgf/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	51.837 kgf/cm ²
App 27-4	= M.A.W.P.	50.328 kgf/cm ²

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
CH. DISH END	1573.3	2386.0	0.659	65.48
CH. SHELL	1631.2	2386.0	0.684	65.48
SHELL	1631.5	2386.0	0.684	65.49
SH. COVER	1573.6	2386.0	0.659	65.49

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
TS IN NOZZLE	Nozzle	1203.20	1203.20	1.000
TS IN NOZZLE	Pad	1407.13	1407.13	1.000
TS OUT NOZZLE	Nozzle	1203.20	1203.20	1.000
TS OUT NOZZLE	Pad	1407.13	1407.13	1.000
SH. IN NOZ-1	Nozzle	1203.20	1203.20	1.000
SH. IN NOZ-1	Pad	1407.13	1407.13	1.000
SH IN NOZ-2	Nozzle	1203.20	1203.20	1.000
SH IN NOZ-2	Pad	1407.13	1407.13	1.000
SH OUT NOZ-1	Nozzle	1203.20	1203.20	1.000
SH OUT NOZ-1	Pad	1407.13	1407.13	1.000
SH OUT NOZ-2	Nozzle	1203.20	1203.20	1.000
SH OUT NOZ-2	Pad	1407.13	1407.13	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
CH. DISH END	1407.13	1407.13	1.000
CH. SHELL	1407.13	1407.13	1.000
CH. FLG1	1407.13	1407.13	1.000
SH. FLG1	1407.13	1407.13	1.000
SHELL	1407.13	1407.13	1.000
SH. COVER	1407.13	1407.13	1.000

Minimum

1.000



Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1203.20	1203.20	1.000
Tubesheet Material	1407.13	1407.13	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
TS IN NOZZLE	680.31	2202.47	0.309
TS OUT NOZZLE	680.31	2202.47	0.309
SH. IN NOZ-1	726.94	2202.47	0.330
SH IN NOZ-2	726.94	2202.47	0.330
SH OUT NOZ-1	680.43	2202.47	0.309
SH OUT NOZ-2	680.43	2202.47	0.309

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: U-TUBESHEET

Shell Data:

Main Shell Description: SHELL

Shell Test Pressure		65.44 kgf/cm ²
Shell Thickness	ts	14.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	520.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1407.13 kgf/cm ²
Shell Allowable Stress at Ambient		1407.13 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Test Pressure		65.43 kgf/cm ²
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	520.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1407.13 kgf/cm ²
Channel Allowable Stress at Ambient		1407.13 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	228
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	3042.00 mm
Straight Tube Length (bet. inner tubsht faces) L		2952.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-210 A-1
Tube Material UNS Number		K02707
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		1203.20 kgf/cm ²
Tube Allowable Stress At Ambient		1203.20 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2600.14 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	23.8120 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	243.630 mm
Largest Center-to-Center Tube Distance	Ul	33.0500 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	6.0000 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-516 70



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K02700
Tubesheet Allowable Stress at Temperature	S	1407.13 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1407.13 kgf/cm ²
Thickness of Tubesheet	h	90.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	595.000 mm
Dimension G for the Channel Side	Gc	571.582 mm
Area of the Untubed Lanes	AL	167.3 cm ²
Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes

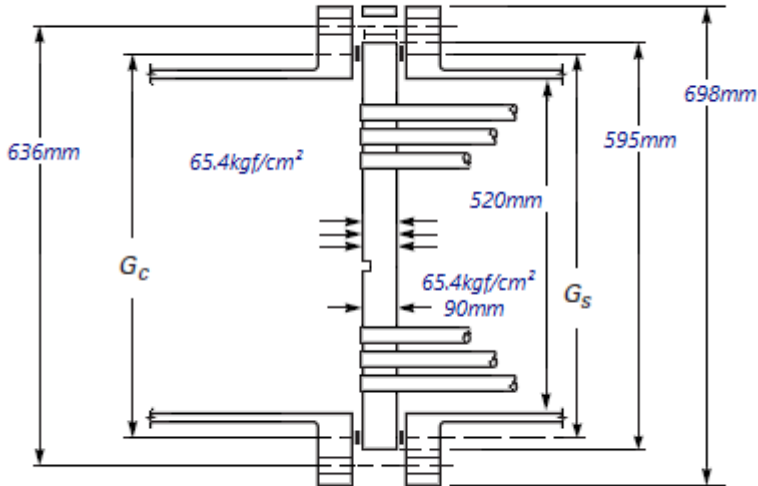
Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	698.000 mm
Flange Inside Diameter	B	520.000 mm
Flange Face Outside Diameter	Fod	598.000 mm
Flange Face Inside Diameter	Fid	520.000 mm
Gasket Outside Diameter	Go	592.000 mm
Gasket Inside Diameter	Gi	526.000 mm
Small end Hub thk.	g0	14.0000 mm
Large end Hub thk.	g1	20.0000 mm
Gasket Factor,	m	3.75
Gasket Design Seating Stress	y	534.33 kgf/cm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	526.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	0.0000
Partition Gasket Design Seating Stress	yPart	0.00 kgf/cm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	636.000 mm
Nominal Bolt Diameter	dB	27.0000 mm
Thread Series		TEMA Metric Thread
Number of Bolts	n	28
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	1753.82 kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82 kgf/cm ²
Weld between Flange and Shell/Channel		0.0000 mm
Alternate Flange Operating Bolt Load, Wm1		258050.92 kgf
Alternate Flange Seating Bolt Load, Wm2		98169.09 kgf
Alternate Flange Design Bolt Load, W		230640.34 kgf
Tubesheet Integral with		None
Tubesheet Extended as Flange		No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (65.43 \cdot (19.05/2 - 0)) / (1203.2 \cdot 1.0 + 0.4 \cdot 65.43)$
 $= 0.5069 + 0.0000 = 0.5069 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 21.11 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	3195.63	9.03	50.0000	0.0134948	1251.46
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 196.1288 \text{ kgf/cm}^2$						

Results for Req'd Thickness for Ext. Pressure (Tca):



TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.8622	19.05	3195.63	22.09	50.0000	0.0022533	1084.49

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 1084.4907) / (3 \cdot 22.0946) = 65.4453 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.8622 mm
Allowable Internal Pressure at Corroded thickness	292.45 kgf/cm ²
Required Internal Design Pressure	65.43 kgf/cm ²
Allowable External Pressure at Corroded thickness	196.13 kgf/cm ²
Required External Design Pressure	65.44 kgf/cm ²
Required Thickness due to Shell Side pressure	0.8622 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (G_{oc} - G_{ic}) / 2$	33.000 mm
Basic Gasket Width,	$b_0 = N / 2$	16.500 mm
Effective Gasket Width,	$b = \sqrt{b_0} \cdot 2.5$	10.232 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	571.536 mm

Warning: Available bolt area, 115.879 cm² for then shell flange is < required 147.137 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 243.63 + 19.05 \\ &= 506.310 \text{ mm} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (23.812 - 19.05) / 23.812 \\ &= 0.200 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$\begin{aligned} &= G_s / D_o \text{ (Configurations d, e, f)} \\ &= 571.5355 / 506.31 \\ &= 1.1288 \end{aligned}$$

Ratio [ρ_c]:

$$\begin{aligned} &= G_c / D_o \text{ (Configurations d)} \\ &= 571.5824 / 506.31 \\ &= 1.1289 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 90.0 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2t((E_t/E) \cdot (S_t/S) \cdot (\rho)); dt - 2t) \\ &= \max(19.05 - 2 \cdot 2.11 (2067025 / 2067025) \cdot \\ &\quad (1203 / 1407) \cdot (0); 19.05 - 2 \cdot 2.11) \end{aligned}$$



= 19.0500 mm

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{(1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2))} \\ &= 23.812 / \sqrt{(1 - 4 \cdot \min(167.33; 4 \cdot 506.31 \cdot 23.812) / (\pi \cdot 506.31^2))} \\ &= 24.8678 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (24.8678 - 19.05) / 24.8678 \\ &= 0.2339 \end{aligned}$$

$h/p = 3.779607$; $\mu^* = 0.233949$; $E^* = 406141 \text{ kgf/cm}^2$

$E^*/E = 0.196486$; $\nu^* = 0.414066$

Note: As h/p (3.780) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / Do \\ &= 595.0 / 506.31 \\ &= 1.1752 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - \nu^*) / E^* \cdot (E \cdot \ln(K)) \\ &= (1 - 0.41) / 406141 \cdot (2067025 \cdot \ln(1.18)) \\ &= 0.4813 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= Do^2 / 16 \cdot [(ps - 1) \cdot (ps^2 + 1) \cdot Ps - (pc - 1) \cdot (pc^2 + 1) \cdot Pt] \\ &= 506.31^2 / 16 \cdot [(1.129 - 1) \cdot (1.129^2 + 1) \cdot 65.438 - \\ &\quad (1.129 - 1) \cdot (1.129^2 + 1) \cdot 65.426] \\ &= -193.3678 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= Mts + W^* \cdot (Gc - Gs) / (2 \cdot \pi \cdot Do) \\ &= -193.4 + 258050.9 \cdot (571.582 - 571.582) / (2 \cdot \pi \cdot 506.31) \\ &= -193.3678 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= (M^*) - Do^2 / 32 \cdot F (Ps - Pt) / (1 + F) \\ &= (-193.37) - 506.31^2 / 32 \cdot 0.481 (65.44 - 65.43) / (1 + 0.48) \\ &= -161.2536 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= Mp + Do^2 / 64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\ &= -161.25 + 506.31^2 / 64 \cdot (3 + 0.414) \cdot (65.44 - 65.43) \\ &= 0.1214 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|Mp|; |Mo|) \\ &= \max(|-161.254|; |0.121|) \\ &= 161.2536 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - hg')^2) \\ &= 6 \cdot 161.254 / (0.2339 (90.0 - 6.0)^2) \\ &= 0.5861 \text{ kgf/cm}^2 \end{aligned}$$



Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot \beta T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 2651.1 \\ &= 5037.13 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 161.291 / (0.2339 \cdot (0.9708 - 6.0)^2) \\ &= 137.1447 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 0.9708 + 0 + 0 \\ &= 0.9708 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(0.9708; 0.0037) \\ &= 0.9708 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |65.44 - 65.43| \\ &= 0.012 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / Do \cdot 0.8 \cdot \beta T \cdot S_y \\ &= 4 \cdot 0.2 \cdot 90.0 / 506.31 \cdot 0.8 \cdot 0.95 \cdot 2651.122 \\ &= 286.499 \text{ kgf/cm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (Do / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + Cats + Catc \\ &= 1 / (4 \cdot 0.2) \cdot (506.31 / 2014.853) \cdot |65.44 - 65.43| + 0 \\ &= 0.0037 \text{ mm} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (19.05 - 2.11) 2470.132 \\ &= 2773.738 \text{ kgf} \end{aligned}$$

Groove Weld Strength [F_g]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 2470.13 \\ &= 2773.7375 \text{ kgf} \end{aligned}$$

Weld Strength Factor [f_w]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 2470.132 / 2470.132 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [f_d]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [F_f]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 2470.13 \end{aligned}$$



= 1842.8876 kgf

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do} \right) \\ = 2 \left(\sqrt{((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 19.05 \right) \\ = 2.562 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm
 Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 2773.74 is >= tube strength 2773.74 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	0.6 <=	5037.1 kgf/cm ²	Ok
Tubesheet shear stress	0.1 <=	2014.9 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	0.971	90.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	74.977	7.643	3387 5037 460 2015	Ps+Pt D1	Ok
D2uc	74.977	7.644	3387 5037 460 2015	Ps+Pt D2	Ok
D3uc	0.971	0.001	1 5037 ... 2015	Ps+Pt D3	Ok
Max:	74.9773	7.644 mm	0.672 0.228 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
----------------	----------	--------	-----



Tubesheet thickness :	74.977	90.000	Ok
Tube thickness :	0.862	2.110	Ok
Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
10	20	No Calc	548	10.825	0.0027436	1124.48
20	30	59.3333	548	10.825	0.0034633	1165.74
30	40	No Calc	...	76.825	No Calc	No Calc
40	50	No Calc	...	76.825	No Calc	No Calc
50	60	314.213	548	10.825	0.00056231	573.25
60	70	No Calc	548	10.825	0.0027436	1124.48

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
10	20	14	No Calc	...	24.6807
20	30	14	No Calc	...	30.7035
30	40	80	68.58	...	No Calc
40	50	80	68.58	...	No Calc
50	60	14	No Calc	...	15.0984
60	70	14	4.5939	1.05461	24.6807

Minimum

15

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	59.3333	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	314.213	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
10	20	53.0802	29024.2	42.5471	29967.7	5.30802
20	30	91.0147	106186	70.7923	108795	9.10147
30	40	128.082	26389.5	123.148	26546.1	12.8082
40	50	128.082	26389.5	123.148	26546.1	12.8082
50	60	554.826	436065	431.549	451971	55.4826
60	70	53.2472	29194.1	42.681	30141.7	5.32472
Total		1008	653248.75	833	673968.19	100



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
10	Liqd	20.5979	-4.33333	0.18167E-05	...	Liquid: 10
20	Liqd	75.3581	25	...	...	Liquid: 20
20	Nozl	55.9501	25	-33.3165	...	TS IN NOZZLE
20	Nozl	54.9726	25	33.3165	...	TS OUT NOZZLE
50	Sadl	54.9535	45	38.3	...	Right Saddle
50	Sadl	54.9535	259.8	38.3	...	right saddle
50	Liqd	8.71738	152.4	...	...	Liquid: 50
50	Nozl	82.9461	85	35.6837	...	SH. IN NOZ-1
50	Nozl	82.9461	219.8	35.6837	...	SH IN NOZ-2
50	Nozl	55.9501	85	-33.3165	...	SH OUT NOZ-1
50	Nozl	55.9501	219.8	-33.3165	...	SH OUT NOZ-2
50	Wght	299.995	152.4	...	...	Baffle
60	Liqd	0.58362	9.41333	0.18167E-05	...	Liquid: 60
30	FTsh	163.482	15.5	...	...	U-TUBESHEET
30	Tube	728.222	163.1	...	...	Tubes

Total Weight of Each Detail Type:

Saddles	109.9
Liquid	105.3
Nozzles	388.7
Weights	300.0
Exchanger Components	891.7
Liquid in Tubes	96.4
Sum of the Detail Weights	1892.0 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1109.2	1109.2	1109.2	1109.2	1109.2	1109.2
Saddles	109.9	109.9	109.9	109.9	109.9	109.9
Nozzles	388.7	388.7	388.7	388.7	388.7	388.7
Wld Weights	300.0	300.0	300.0	300.0	300.0	300.0
Exchanger	891.7	891.7	891.7	891.7	891.7	891.7
Ope. Liquid	...	...	...	...	...	105.3
Tube Ope Lqd	...	...	...	...	...	96.4
Test Liquid	...	653.0	...	...	...	...
Tube Tst Lqd	...	135.8	...	...	...	...
Totals	2799.5	3588.3	2799.5	2799.5	2799.5	3001.2



Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2799.5 kg
Shop Test Wt.	- Fabricated weight + water (full)	3588.3 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	2799.5 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	2799.5 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	2799.5 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	3001.2 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	2809.3 kg
Field Test Wt.	- Empty weight + test medium (full)	3417.1 kg

Exchanger Tube Data

Volume of Exchanger tubes :	135896.7 cm ³
Weight of Ope Liq in tubes :	96.4 kg
Weight of Water in tubes :	135.8 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	4170.03	3747.91
20	30	8607.96	8168.14
30	40	4247.97	1993.03
40	50	4247.97	1993.03
50	60	52474.1	49793
60	70	4183.9	3760.98

Total	77932.0	69456.1 cm ²
Total	83.9	74.8 ft ²

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Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
TS IN NOZZLE	99.60	104.01	66	600	GR 1.1	...	...	...	...
TS OUT NOZZLE	99.60	104.01	66	600	GR 1.1	...	...	...	...
SH. IN NOZ-1	96.79	104.01	88	600	GR 1.1	...	...	...	...
SH IN NOZ-2	96.79	104.01	88	600	GR 1.1	...	...	...	...
SH OUT NOZ-1	96.79	104.01	88	600	GR 1.1	...	...	...	...
SH OUT NOZ-2	96.79	104.01	88	600	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 96.790 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 104.014 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 99.600 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 104.014 kgf/cm²

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 7.695

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)

= 1983.5 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.1 • 0.854 • 1983.503

= 160.7 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
10	0.83	1.0000	1.0000	1.0000	57.50	0.8539
20	0.83	1.0000	1.0000	1.0000	57.50	0.8539
30	0.83	1.0000	1.0000	1.0000	57.50	0.8539
40	0.83	1.0000	1.0000	1.0000	57.50	0.8539
50	0.83	1.0000	1.0000	1.0000	57.50	0.8539
60	0.83	1.0000	1.0000	1.0000	57.50	0.8539

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation cm	Pressure kgf/m ²	Force kgf
Baffle	0.00	83.00	202.34	0.00

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm ²	Wind Pressure kgf/m ²	Element Wind Load kgf
10	20	83	65.76	948.573	202.344	18.0222
20	30	83	65.76	3288	202.344	62.4697
30	40	83	62.4	686.4	202.344	13.0411
40	50	83	62.4	686.4	202.344	13.0411
50	60	83	65.76	20043.6	202.344	380.815



60	70	83	65.76	953.834	202.344	18.1222
----	----	----	-------	---------	---------	---------

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
= (Sa/g)ss • DampVal) • (I / R)
= (0.0939 • 1.0) • 1.0
= 0.094

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
= Earthquake Weight • Ah • Scalar
= 337.42 • 0.0939 • 1.0
= 31.7 kgf

Vertical load per 8.4:
= 2/3 • Ah • W • Scalar
= 2/3 • 0.0939 • 2699 • 1.0
= 168.9 kgf

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------

From	To	Height cm	Weight kgf	Ope Load kgf
10	20	26	337.421	31.669
20	30	26	337.421	31.669
30	40	26	337.421	31.669
40	50	26	337.421	31.669
50	Sad1	26	337.421	31.669
Sad1	60	26	337.421	31.669
50	60	26	337.421	31.669
60	70	26	337.421	31.669

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	238.718 cm
Center of Gravity of the Liquid	42.622 cm
Center of Gravity of the Nozzles	179.158 cm
Center of Gravity of the Added Weights (Operating)	238.718 cm
Center of Gravity of the Added Weights (Empty)	238.718 cm
Center of Gravity of the Tubesheet(s)	70.500 cm
Center of Gravity of the Tubes	218.100 cm
Center of Gravity of Bare Shell New and Cold	172.971 cm
Center of Gravity of Bare Shell Corroded	167.285 cm
Vessel CG in the Operating Condition	183.964 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	189.212 cm
Vessel CG in the Test Condition	190.022 cm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

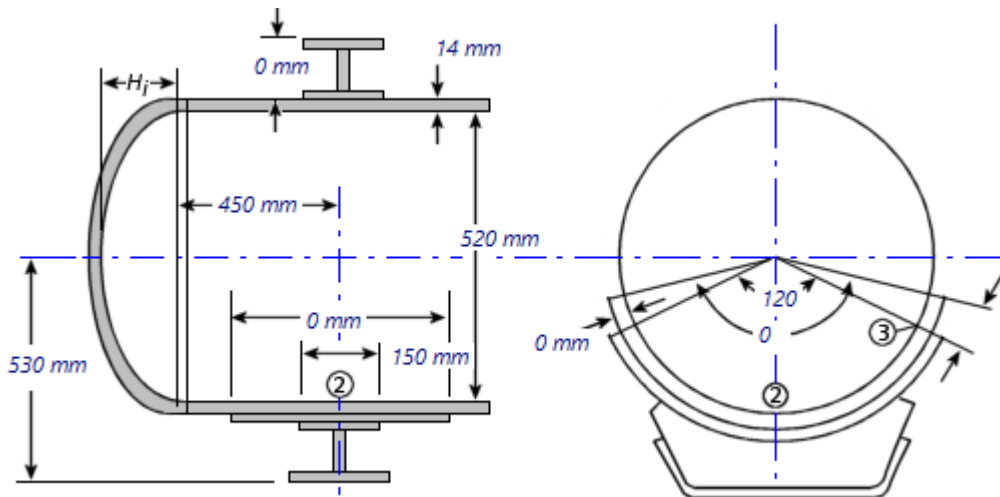
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{268.5875 \cdot 10.825}, 2 \cdot 450.0)$$

$$= 234.1165 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	268.59	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	320.88	cm
Distance from Saddle to Vessel tangent	a1 or a	450.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	530.00	mm
Coefficient of Friction	mu	0.50	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		2967.22	kgf
Pressure used in Analysis	P	47.201	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	541.61	1407.13
Long. Stress at Bottom of Midspan	629.52	1407.13
Long. Stress at Top of Saddles	644.41	1407.13
Long. Stress at Bottom of Saddles	552.95	1407.13
Tangential Shear in Shell	85.97	1125.71
Circ. Stress at Horn of Saddle	230.02	1758.92
Circ. Compressive Stress in Shell	8.90	1407.13



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(505.5/2 + 0) \cdot 530.0/491.9024$
 $= 817.0\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(75.6, 0) \cdot 530.0/2148.0002$
 $= 18.7\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(253.35, 0) \cdot 530.0/2148.0002$
 $= 62.5\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(253/2 + 0) \cdot 530.0/491.9024$
 $= 409.5\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 2150 + \max(19, 817, 63, 409)$
 $= 2967.2\ kgf$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1984.306(0.0 + \pi/4(0.548 \cdot 1.2)^2)$
 $= 741.341\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3022.18	kgf
Transverse Shear Load	252.76	kgf
Longitudinal Shear Load	253.35	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(520.0/2 + 14.0 + 10.0) \sin(120.0/2), 2 \cdot 268.587)$
 $= 491.902\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -2967 \cdot 45.0 [1 - (1 - 45.0/320.88 + (26.859^2 - 0.0^2) / (2 \cdot 45.0 \cdot 320.88)) / (1 + (4 \cdot 0.0) / (3 \cdot 320.88))]$
 $= -153.9\ kgf \cdot m$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$
 $= 2967 \cdot 321 / 4 ((1 + 2(27^2 - 0^2)/(321^2)) / (1 + (4 \cdot 0) / (3 \cdot 321)) - 4 \cdot 45 / 321)$
 $= 1078.4\ kgf \cdot m$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$
 $= 47.201 \cdot 268.587 / (2 \cdot 10.825) - 1078.4 / (\pi \cdot 268.6^2 \cdot 10.825)$
 $= 541.61\ kgf/cm^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 47.201 \cdot 268.587 / (2 \cdot 10.825) + 1078.4 / (\pi \cdot 268.6^2 \cdot 10.825)$$

$$= 629.52 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 47.201 \cdot 268.587 / (2 \cdot 10.825) - 153.9 / (0.1066 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 644.41 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 47.201 \cdot 268.587 / (2 \cdot 10.825) + 153.9 / (0.1923 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 552.95 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm / 3))$$

$$= 2967(320.88 - 2 \cdot 45.0) / (320.88 + (4 \cdot 0.0 / 3))$$

$$= 2135.0 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 2134.98 / (268.5875 \cdot 10.825)$$

$$= 85.97 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(268.587 \cdot 10.825)}$$

$$= 42.058 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [$\sigma6$]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 2967 \cdot 0.1 / (10.825(150.0 + 42.06 + 42.06))$$

$$= -8.90 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [$\sigma7$]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -2967 / (4 \cdot 10.825(150.0 + 42.058 + 42.058)) -$$

$$3 \cdot 0.0529 \cdot 2967 / (2 \cdot 10.825^2)$$

$$= -230.02 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(268.587 \cdot 10.825)}, 2 \cdot 45.0)$$

$$= 234.12 \text{ mm}$$

Distance between Saddle Supports [Ls]:

$$= 2148.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 2148.0(87.8 - 21.1)$$

$$= 1.722 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	500.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2461.9	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$



$$= ((1.56 \cdot \sqrt{(263.175 \cdot 10.825)} + 225.0) 10.825^3) / 12$$

$$= 3.259 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 3.259 + 33.37 (97.676 - 5.413)^2$$

$$= 2843.888 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	308.3	10.8	5.4	33.4	18061.3	0.284E+04
Wearplate	225.0	10.0	15.8	22.5	35606.2	0.151E+04
Web	12.0	230.0	135.8	27.6	374876.9	0.162E+04
BasePlate	150.0	16.0	258.8	24.0	621179.9	0.624E+04
Totals	...	...	...	107.5	1049724.4	0.122E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1049724 / 107.47$$

$$= 97.676 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2967.222$$

$$= 603.8940 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 8.1497 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1477.1587 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 530.0 - 263.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 312.356 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 188.6302 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 15.0908 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1641.2876 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(2967 + 55)150.0 / (2 \cdot 500.0 \cdot 1641.288))}$$

$$= 9.103 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (260.0 + 14.0 + 10.0)$$

$$= 491.902 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 491.9024 / (3 - 1)$$

$$= 245.951 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 245.9512 \cdot 150.0 / 2$$



$$= 184.463 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3022.176 / (500.0 \cdot 150.0)$$

$$= 4.030 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 184.5 \cdot 4.03$$

$$= 743.308 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 16.56 + 14.757$$

$$= 31.317 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 743.3 / 31.3171$$

$$= 23.735 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	470.
Web	123.0	12.0	6.0	14.8	8854.2	234.
Totals	...	...	...	31.3	142990.2	704.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 12.0$$

$$= 138.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 142990.234 / 31.317$$

$$= 45.659 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(45.659, 150.0 - 45.659)$$

$$= 104.341 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(703.5 / 31.317)}$$

$$= 47.396 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 530.0 - \cos(120.0/2) (260.0 + 14.0 + 10.0) - 16.0$$

$$= 372.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2028094 / 2461.9)}$$

$$= 127.518$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 372.0 / 47.396$$

$$= 7.849$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 253.4 / (2 \cdot 500.0) \cdot 245.951 \cdot 372.0 / 2$$



= 11.590 kgf-m.

Compressive Allowable, $KL/r < Cc$ (7.8487 < 127.5179) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (7.85)^2 / (2 \cdot 127.52^2)) 2462 / (5/3 + 3 \cdot (7.85) / (8 \cdot 127.52) - (7.85^3) / (8 \cdot 127.52^3))$$

$$= 1454 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 23.73/1454.25 + (11.59 \cdot 104.341/7035148) / 1641.29$$

$$= 0.027$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	645.
Web	246.0	12.0	6.0	29.5	17708.5	218.
Totals	...	...	...	46.1	151844.5	863.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 151844.469/46.074$$

$$= 32.957 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(32.957, \text{abs}(150.0 - 32.957))$$

$$= 117.043 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - Bpthk$$

$$= 530.0 - \cos(284.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 230.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (863.0/46.074) }$$

$$= 43.280 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 230.0/43.28$$

$$= 5.314$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 253.352 / (2 \cdot 500.0)$$

$$= 0.253 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.253 \cdot 245.951 \cdot 230.0$$

$$= 14.332 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 14.332/73737.445$$

$$= 19.436 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < Cc$ (5.3142 < 127.5179) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (5.31)^2 / (2 \cdot 127.52^2)) 2462 / (5/3 + 3 \cdot (5.31) / (8 \cdot 127.52) - (5.31^3) / (8 \cdot 127.52^3))$$

$$= 1462 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1 / I) / Sba$$



$$= 34.55/1462.17 + (14.33 \cdot 117.043/863.048)/1641.29$$

$$= 0.035$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	20.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BlArea	1.4584 cm ²
Saddle Load QO (Weight)	QO	2205.2 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	62.5 kgf
Maximum Transverse Force	Ft	252.8 kgf
Maximum Longitudinal Force	Fl	253.4 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_{b,l}]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 253.4 / (1.458 \cdot 2)$$

$$= 86.9 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_{b,t}]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 252.8 / (1.458 \cdot 2)$$

$$= 86.7 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 253.35 / (1101.26 \cdot 2.0)$$

$$= 0.1150 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 530.0 \cdot 252.76 + 0.0$$

$$= 133.96 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 133.96/2205.18$$

$$= 60.75 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.115, 0.0]$$

$$= 0.1150 \text{ cm}^2$$

MDMT Calculations for: Right Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, c = 3.175 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.889, Temp. Reduction = 6 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B

-29 °C



Min Metal Temp. at Required thickness (UCS 66.1) -35 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{ 268.5875 \cdot 10.825 }, 2 \cdot 450.0)$$

$$= 234.1165 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	268.59	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	320.88	cm
Distance from Saddle to Vessel tangent	a1 or a	450.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	13.86	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	530.00	mm
Coefficient of Friction	mu	0.50	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		1535.03	kgf
Pressure used in Analysis	P	47.201	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	565.77	1407.13
Long. Stress at Bottom of Midspan	605.36	1407.13
Long. Stress at Top of Saddles	630.39	1407.13
Long. Stress at Bottom of Saddles	560.72	1407.13
Tangential Shear in Shell	42.05	1125.71
Circ. Stress at Horn of Saddle	118.99	1758.92
Circ. Compressive Stress in Shell	4.60	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(505.5/2 + 0) \cdot 530.0/491.9024$$

$$= 817.0 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(75.6, 0) \cdot 530.0/2148.0002$$

$$= 18.7 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(253.35, 0) \cdot 530.0/2148.0002$$

$$= 62.5 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$



$$= 3.0(253/2 + 0) \cdot 530.0/491.9024$$

$$= 409.5 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 718 + \max(19, 817, 63, 409)$$

$$= 1535.0 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.548 \cdot 1.2)^2)$$

$$= 741.341 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1589.99	kgf
Transverse Shear Load	252.76	kgf
Longitudinal Shear Load	253.35	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(520.0/2 + 14.0 + 10.0)\sin(120.0/2), 2 \cdot 268.587)$$

$$= 491.902 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -1535 \cdot 45.0 [1 - (1 - 45.0/320.88 + (26.859^2 - 13.859^2)/(2 \cdot 45.0 \cdot 320.88))/(1 + (4 \cdot 13.86)/(3 \cdot 320.88))]$$

$$= -117.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 1535 \cdot 321/4((1 + 2(27^2 - 14^2)/(321^2))/(1 + (4 \cdot 14)/(3 \cdot 321)) - 4 \cdot 45/321)$$

$$= 485.6 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 47.201 \cdot 268.587/(2 \cdot 10.825) - 485.6/(\pi \cdot 268.6^2 \cdot 10.825)$$

$$= 565.77 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$

$$= 47.201 \cdot 268.587/(2 \cdot 10.825) + 485.6/(\pi \cdot 268.6^2 \cdot 10.825)$$

$$= 605.36 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 47.201 \cdot 268.587/(2 \cdot 10.825) - 117.2/(0.1066 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 630.39 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M_1/(K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 47.201 \cdot 268.587/(2 \cdot 10.825) + 117.2/(0.1923 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 560.72 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 1535(320.88 - 2 \cdot 45.0)/(320.88 + (4 \cdot 13.86/3))$$

$$= 1044.3 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (Rm \cdot t) \\
 &= 1.1707 \cdot 1044.35 / (268.5875 \cdot 10.825) \\
 &= 42.05 \text{ kgf/cm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x1,x2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{ (Rm \cdot t) } \\
 &= 0.78 \cdot \sqrt{ (268.587 \cdot 10.825) } \\
 &= 42.058 \text{ mm}
 \end{aligned}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (t (b + X1 + X2)) \\
 &= -0.7603 \cdot 1535 \cdot 0.1 / (10.825 (150.0 + 42.06 + 42.06)) \\
 &= -4.60 \text{ kgf/cm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$\begin{aligned}
 &= -Q / (4 \cdot t (b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2) \\
 &= -1535 / (4 \cdot 10.825 (150.0 + 42.058 + 42.058)) - \\
 &\quad 3 \cdot 0.0529 \cdot 1535 / (2 \cdot 10.825^2) \\
 &= -118.99 \text{ kgf/cm}^2
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min (b + 1.56 \cdot \sqrt{ (Rm \cdot t) }, 2a) \\
 &= \min (150.0 + 1.56 \cdot \sqrt{ (268.587 \cdot 10.825) }, 2 \cdot 45.0) \\
 &= 234.12 \text{ mm}
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	500.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2461.9	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12 \\
 &= ((1.56 \cdot \sqrt{ (263.175 \cdot 10.825) } + 225.0) 10.825^3) / 12 \\
 &= 3.259 \text{ cm}^4
 \end{aligned}$$

Resolved Inertia for the Shell [I,shell]:

$$\begin{aligned}
 &= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2 \\
 &= 3.259 + 33.37 (97.676 - 5.413)^2 \\
 &= 2843.888 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	308.3	10.8	5.4	33.4	18061.3	0.284E+04
Wearplate	225.0	10.0	15.8	22.5	35606.2	0.151E+04
Web	12.0	230.0	135.8	27.6	374876.9	0.162E+04
BasePlate	150.0	16.0	258.8	24.0	621179.9	0.624E+04
Totals	...	...	...	107.5	1049724.4	0.122E+05

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 1049724 / 107.47 \\
 &= 97.676 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 120.0 / 2 \\
 &= 120.0
 \end{aligned}$$



Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 1535.035$$

$$= 312.4128 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.2161 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1477.1587 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 530.0 - 263.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 312.356 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 97.5841 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 7.8070 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1641.2876 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1535 + 55) \cdot 150.0) / (2 \cdot 500.0 \cdot 1641.288)}$$

$$= 6.603 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (260.0 + 14.0 + 10.0)$$

$$= 491.902 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 491.9024 / (3 - 1)$$

$$= 245.951 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 245.9512 \cdot 150.0 / 2$$

$$= 184.463 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1589.988 / (500.0 \cdot 150.0)$$

$$= 2.120 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 184.5 \cdot 2.12$$

$$= 391.059 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 16.56 + 14.757$$

$$= 31.317 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 391.1 / 31.3171$$

$$= 12.487 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

B

D

Y

A

AY

Io



Rib	12.0	138.0	81.0	16.6	134136.0	470.
Web	123.0	12.0	6.0	14.8	8854.2	234.
Totals	...	...	...	31.3	142990.2	704.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 150.0 - 12.0
 = 138.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 142990.234/31.317
 = 45.659 mm

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(45.659, 150.0 - 45.659)
 = 104.341 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{703.5/31.317}$
 = 47.396 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 530.0 - cos(120.0/2)(260.0 + 14.0 + 10.0) - 16.0
 = 372.000 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2028094/2461.9}$
 = 127.518

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 372.0/47.396
 = 7.849

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 253.4 / (2 • 500.0) • 245.951 • 372.0/2
 = 11.590 kgf-m.

Compressive Allowable, KL/r < Cc (7.8487 < 127.5179) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (7.85)^2 / (2 \cdot 127.52^2)) 2462 / (5/3 + 3 \cdot (7.85) / (8 \cdot 127.52) - (7.85^3) / (8 \cdot 127.52^3))$
 = 1454 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 12.49/1454.25 + (11.59 • 104.341/7035148) / 1641.29
 = 0.019

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	645.
Web	246.0	12.0	6.0	29.5	17708.5	218.
Totals	...	...	...	46.1	151844.5	863.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 151844.469/46.074
 = 32.957 mm



Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(32.957, \text{abs}(150.0 - 32.957))$$

$$= 117.043 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 530.0 - \cos(284.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 230.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (863.0/46.074) }$$

$$= 43.280 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 230.0/43.28$$

$$= 5.314$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 253.352 / (2 \cdot 500.0)$$

$$= 0.253 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.253 \cdot 245.951 \cdot 230.0$$

$$= 14.332 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 14.332/73737.445$$

$$= 19.436 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ ($5.3142 < 127.5179$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.31)^2 / (2 \cdot 127.52^2)) 2462 /$$

$$(5/3 + 3 \cdot (5.31) / (8 \cdot 127.52) - (5.31^3) / (8 \cdot 127.52^3))$$

$$= 1462 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 18.18/1462.17 + (14.33 \cdot 117.043/863.048) / 1641.29$$

$$= 0.024$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	20.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	1.4584 cm ²
Saddle Load Q0 (Weight)	Q0	773.0 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	62.5 kgf
Maximum Transverse Force	Ft	252.8 kgf
Maximum Longitudinal Force	Fl	253.4 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [$\tau_{b,l}$]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 253.4 / (1.458 \cdot 2)$$

$$= 86.9 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$



Shear Stress in a Single Bolt, Transverse Direction [Tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $252.8 / (1.458 \cdot 2)$
= 86.7 kgf/cm^2 . Must be less than 1101.3 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
= $253.35 / (1101.26 \cdot 2.0)$
= 0.1150 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $530.0 \cdot 252.76 + 0.0$
= 133.96 kgf-m .

Eccentricity (e):
= R_{mom} / QO
= $133.96 / 772.99$
= $173.30 \text{ mm} > B_{plen} / 6$ --> Uplift in Transverse direction

$f = B_{plen} / 2 - \text{Edgedis}$
= $500.0 / 2 - 50.0$
= 200.00 mm

Modular Ratio Of Steel/Concrete (n1):
= E_S / E_C
= $2074057 / 219499.77$
= 9.45

$K1 = 3 (e - 0.5 \cdot B_{plen})$
= $3 (173.3 - 0.5 \cdot 500.0)$
= -230.10 mm

$K2 = 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$
= $6 \cdot 9.45 \cdot 1.46 / 150.0 \cdot (200.0 + 173.3)$
= 20577.13 mm^2

$K3 = -K2 \cdot (0.5 \cdot B_{plen} + f)$
= $-20577.13 \cdot (0.5 \cdot 500.0 + 200.0)$
= -9259710.79 mm^3

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -230.1 \cdot Y^2 + 20577.13 \cdot Y + -9259711.0 = 0$
 $Y = 276.68 \text{ mm}$

$\text{Num} = (B_{plen} / 2 - Y / 3 - e)$
= $(500.0 / 2 - 276.68 / 3 - 173.3)$
= -15.53

$\text{Denom} = (B_{plen} / 2 - Y / 3 + f)$
= $(500.0 / 2 - 276.68 / 3 + 200.0)$
= 357.77

Total Bolt Tension Force [Tforce]:
= $-QO \cdot \text{Num} / \text{Denom}$
= $-772.99 \cdot -15.53 / 357.77$
= 33.55 kgf

Bolt Area Required due to Transverse Load [Bltareart]:
= $T_{force} / (S_{tba} \cdot N_{bt})$
= $33.55 / (1396.96 \cdot 1.0)$
= 0.0240 cm^2



Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.115, 0.024]
= 0.1150 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
= 2(QO + Tforce) / (Y • Bpwid)
= 2(772.99 + 33.55)/(276.68 • 150.0)
= 3.89 kgf/cm²

Distance from Baseplate Edge to the Web [ADIST]:
= (Bplen - WebIngh) / 2
= (500.0 - 449.2)/2
= 25.4000 mm

Overturning Moment due To Bolt Tension [Mt]:
= Tforce • Adist
= 33.55 • 2.54
= 0.85 kgf-m.

Equivalent Bearing Pressure (f1):
= fc • (Y - Adist) / Y
= 3.89 • (276.68 - 25.4)/276.68
= 3.53 kgf/cm²

Overturning Moment due to Bearing Pressure [Mc]:
= (Adist² • Bpwid / 6) • (f1 + 2 • fc)
= (25.4² • 150.0/6) • (3.53 + 2 • 3.89)
= 1.82 kgf-m.

Baseplate Required Thickness [Treq]:
= $\sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))}$
= $\sqrt{(6 \cdot \max(0.85, 1.82) / (150.0 \cdot 2110.7))}$
= 1.8588 mm

MDMT Calculations for: right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 10.0, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr • E*/(tg - c) = 0.889, Temp. Reduction = 6 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -35 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

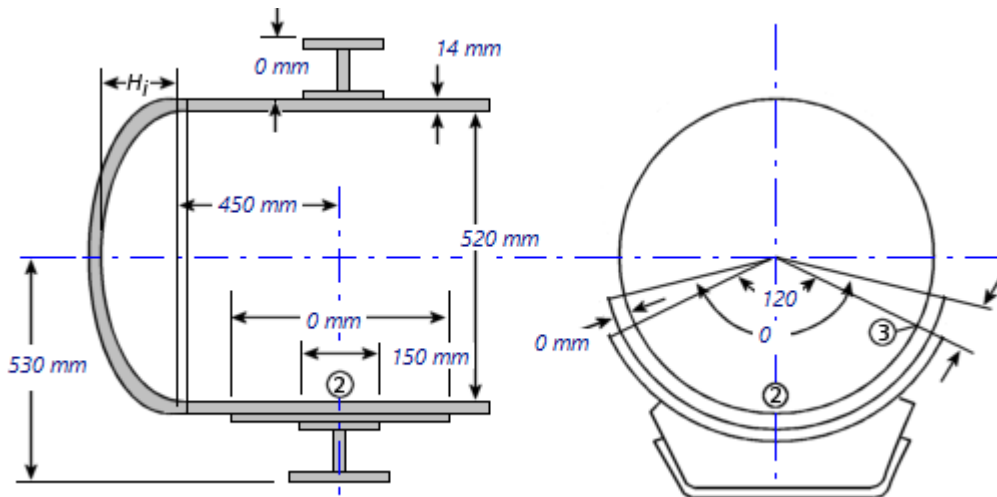
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{268.5875 \cdot 10.825}, 2 \cdot 450.0)$$

$$= 234.1165 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	268.59	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	320.88	cm
Distance from Saddle to Vessel tangent	a1 or a	450.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	530.00	mm
Coefficient of Friction	mu	0.50	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2822.25	kgf
Pressure used in Analysis	P	65.465	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	770.34	2386.01
Long. Stress at Bottom of Midspan	853.96	2386.01
Long. Stress at Top of Saddles	868.12	2386.01
Long. Stress at Bottom of Saddles	781.13	2386.01
Tangential Shear in Shell	81.77	1413.05
Circ. Stress at Horn of Saddle	218.78	3579.01
Circ. Compressive Stress in Shell	8.47	2386.01



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (166.8/2 + 0) \cdot 530.0/491.9024$
 $= 269.6 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(24.95, 0) \cdot 530.0/2148.0002$
 $= 6.2 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 2553 + \max(6, 270, 0, 0)$
 $= 2822.3 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.548 \cdot 1.2)^2)$
 $= 741.341 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2877.21	kgf
Transverse Shear Load	83.41	kgf
Longitudinal Shear Load	24.95	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(520.0/2 + 14.0 + 10.0) \sin(120.0/2), 2 \cdot 268.587)$
 $= 491.902 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -2822 \cdot 45.0 [1 - (1 - 45.0/320.88 + (26.859^2 - 0.0^2)/(2 \cdot 45.0 \cdot 320.88))/(1 + (4 \cdot 0.0)/(3 \cdot 320.88))]$
 $= -146.4 \text{ kgf-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 2822 \cdot 321/4 ((1 + 2(27^2 - 0^2)/(321^2))/(1 + (4 \cdot 0)/(3 \cdot 321)) - 4 \cdot 45/321)$
 $= 1025.7 \text{ kgf-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 65.465 \cdot 268.587/(2 \cdot 10.825) - 1025.7/(\pi \cdot 268.6^2 \cdot 10.825)$
 $= 770.34 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 65.465 \cdot 268.587/(2 \cdot 10.825) + 1025.7/(\pi \cdot 268.6^2 \cdot 10.825)$
 $= 853.96 \text{ kgf/cm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 \cdot t)$
 $= 65.465 \cdot 268.587/(2 \cdot 10.825) - 146.4/(0.1066 \cdot \pi \cdot 268.6^2 \cdot 10.825)$
 $= 868.12 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:



$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 65.465 \cdot 268.587 / (2 \cdot 10.825) + 146.4 / (0.1923 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 781.13 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 2822(320.88 - 2 \cdot 45.0) / (320.88 + (4 \cdot 0.0/3))$$

$$= 2030.7 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 2030.67 / (268.5875 \cdot 10.825)$$

$$= 81.77 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(268.587 \cdot 10.825)}$$

$$= 42.058 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 2822 \cdot 0.1 / (10.825(150.0 + 42.06 + 42.06))$$

$$= -8.47 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -2822 / (4 \cdot 10.825(150.0 + 42.058 + 42.058)) - 3 \cdot 0.0529 \cdot 2822 / (2 \cdot 10.825^2)$$

$$= -218.78 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(268.587 \cdot 10.825)}, 2 \cdot 45.0)$$

$$= 234.12 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	500.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(263.175 \cdot 10.825)} + 225.0)^3) / 12$$

$$= 3.259 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 3.259 + 33.37(97.676 - 5.413)^2$$

$$= 2843.888 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	308.3	10.8	5.4	33.4	18061.3	0.284E+04
Wearplate	225.0	10.0	15.8	22.5	35606.2	0.151E+04
Web	12.0	230.0	135.8	27.6	374876.9	0.162E+04
BasePlate	150.0	16.0	258.8	24.0	621179.9	0.624E+04
Totals	...	...	...	107.5	1049724.4	0.122E+05

Distance to Centroid [C1]:

$$= AY / A$$



$$= 1049724/107.47$$
$$= 97.676 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$
$$= 180 - 120.0/2$$
$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$
$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$
$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$
$$= 0.204 \cdot 2822.253$$
$$= 574.3898 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 7.7516 \text{ kgf/cm}^2$$
$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$
$$= 530.0 - 263.17 \cdot \sin(120.0/2) / 1.0472$$
$$= 312.356 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 179.4143 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 14.3535 \text{ kgf/cm}^2$$
$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt[3]{(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$
$$= \sqrt[3]{(3(2822 + 55)150.0 / (2 \cdot 500.0 \cdot 1767.415))}$$
$$= 8.559 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$
$$= 2 \cdot \cos(90 - 120.0/2) (260.0 + 14.0 + 10.0)$$
$$= 491.902 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$
$$= 491.9024 / (3 - 1)$$
$$= 245.951 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$
$$= 245.9512 \cdot 150.0/2$$
$$= 184.463 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$
$$= 2877.207 / (500.0 \cdot 150.0)$$
$$= 3.836 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$
$$= 184.5 \cdot 3.84$$
$$= 707.652 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$
$$= 16.56 + 14.757$$
$$= 31.317 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$



= 707.7/31.3171
 = 22.596 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	470.
Web	123.0	12.0	6.0	14.8	8854.2	234.
Totals	...	...	...	31.3	142990.2	704.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 150.0 - 12.0
 = 138.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 142990.234/31.317
 = 45.659 mm

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(45.659, 150.0 - 45.659)
 = 104.341 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{703.5/31.317}$
 = 47.396 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 530.0 - cos(120.0/2)(260.0 + 14.0 + 10.0) - 16.0
 = 372.000 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2028094/2651.1}$
 = 122.884

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 372.0/47.396
 = 7.849

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 24.9/ (2 • 500.0) • 245.951 • 372.0/2
 = 1.141 kgf-m.

Compressive Allowable, KL/r < Cc (7.8487 < 122.8836) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (7.85)^2 / (2 \cdot 122.88^2)) 2651 / (5/3 + 3 \cdot (7.85) / (8 \cdot 122.88) - (7.85^3) / (8 \cdot 122.88^3))$
 = 1565 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 22.6/1564.97 + (1.14 • 104.341/7035148) / 1767.42
 = 0.015

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	645.



Web	246.0	12.0	6.0	29.5	17708.5	218.
Totals	...	...	...	46.1	151844.5	863.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 151844.469/46.074$$

$$= 32.957 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(32.957, \text{abs}(150.0 - 32.957))$$

$$= 117.043 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 530.0 - \cos(284.0/2)(16.0 + 0.0 + 0.0) - 0.0$$

$$= 230.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 863.0/46.074 }$$

$$= 43.280 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 230.0/43.28$$

$$= 5.314$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 24.947/(2 \cdot 500.0)$$

$$= 0.025 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.025 \cdot 245.951 \cdot 230.0$$

$$= 1.411 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1.411/73737.445$$

$$= 1.914 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (5.3142 < 122.8836) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.31)^2 / (2 \cdot 122.88^2)) 2651 /$$

$$(5/3 + 3 \cdot (5.31) / (8 \cdot 122.88) - (5.31^3) / (8 \cdot 122.88^3))$$

$$= 1574 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 32.89/1573.88 + (1.41 \cdot 117.043/863.048) / 1767.42$$

$$= 0.022$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	20.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	1.4584 cm ²
Saddle Load Q0 (Weight)	Q0	2607.6 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	6.2 kgf
Maximum Transverse Force	Ft	83.4 kgf
Maximum Longitudinal Force	Fl	24.9 kgf



Saddle Bolted to Steel Foundation

No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $24.9 / (1.458 \cdot 2)$
 = 8.6 kgf/cm^2 . Must be less than 1052.3 kgf/cm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $83.4 / (1.458 \cdot 2)$
 = 28.6 kgf/cm^2 . Must be less than 1052.3 kgf/cm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolls})$
 = $24.95 / (1052.29 \cdot 2.0)$
 = 0.0119 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $530.0 \cdot 83.41 + 0.0$
 = 44.21 kgf-m .

Eccentricity (e):
 = R_{mom} / QO
 = $44.21 / 2607.6$
 = $16.95 \text{ mm} < B_{plen} / 6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.0119, 0.0]$
 = 0.0119 cm^2

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 = $\min(150.0 + 1.56 \cdot \sqrt{268.5875 \cdot 10.825}, 2 \cdot 450.0)$
 = 234.1165 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	268.59	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	320.88	cm
Distance from Saddle to Vessel tangent	a1 or a	450.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	13.86	cm
Shell Allowable Stress used in Calculation		2386.01	kgf/cm ²
Head Allowable Stress used in Calculation		2386.01	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	



Distance from Saddle Base to Centerline B 530.00 mm
 Coefficient of Friction mu 0.50

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 1195.33 kgf
 Pressure used in Analysis P 65.465 kgf/cm²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	796.74	2386.01
Long. Stress at Bottom of Midspan	827.56	2386.01
Long. Stress at Top of Saddles	847.05	2386.01
Long. Stress at Bottom of Saddles	792.80	2386.01
Tangential Shear in Shell	32.75	1413.05
Circ. Stress at Horn of Saddle	92.66	3579.01
Circ. Compressive Stress in Shell	3.59	2386.01

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (166.8/2 + 0) \cdot 530.0/491.9024$
 $= 269.6 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(24.95, 0) \cdot 530.0/2148.0002$
 $= 6.2 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 926 + \max(6, 270, 0, 0)$
 $= 1195.3 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.548 \cdot 1.2)^2)$
 $= 741.341 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 1250.29 kgf
 Transverse Shear Load 83.41 kgf
 Longitudinal Shear Load 24.95 kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(520.0/2 + 14.0 + 10.0) \sin(120.0/2), 2 \cdot 268.587)$
 $= 491.902 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h m^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$
 $= -1195 \cdot 45.0 [1 - (1 - 45.0/320.88 + (26.859^2 - 13.859^2)/(2 \cdot 45.0 \cdot 320.88)) / (1 + (4 \cdot 13.86)/(3 \cdot 320.88))]$
 $= -91.3 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:



$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L)$$

$$= 1195 \cdot 321 / 4 \cdot ((1 + 2(27^2 - 14^2) / (321^2)) / (1 + (4 \cdot 14) / (3 \cdot 321)) - 4 \cdot 45 / 321)$$

$$= 378.1 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 65.465 \cdot 268.587 / (2 \cdot 10.825) - 378.1 / (\pi \cdot 268.6^2 \cdot 10.825)$$

$$= 796.74 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 65.465 \cdot 268.587 / (2 \cdot 10.825) + 378.1 / (\pi \cdot 268.6^2 \cdot 10.825)$$

$$= 827.56 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 65.465 \cdot 268.587 / (2 \cdot 10.825) - 91.3 / (0.1066 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 847.05 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 65.465 \cdot 268.587 / (2 \cdot 10.825) + 91.3 / (0.1923 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 792.80 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 1195(320.88 - 2 \cdot 45.0) / (320.88 + (4 \cdot 13.86 / 3))$$

$$= 813.2 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 813.24 / (268.587 \cdot 10.825)$$

$$= 32.75 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(268.587 \cdot 10.825)}$$

$$= 42.058 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 1195 \cdot 0.1 / (10.825(150.0 + 42.06 + 42.06))$$

$$= -3.59 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -1195 / (4 \cdot 10.825(150.0 + 42.058 + 42.058)) - 3 \cdot 0.0529 \cdot 1195 / (2 \cdot 10.825^2)$$

$$= -92.66 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(268.587 \cdot 10.825)}, 2 \cdot 45.0)$$

$$= 234.12 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	500.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)



Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(263.175 \cdot 10.825)} + 225.0)^3) / 12$$

$$= 3.259 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 3.259 + 33.37 (97.676 - 5.413)^2$$

$$= 2843.888 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	308.3	10.8	5.4	33.4	18061.3	0.284E+04
Wearplate	225.0	10.0	15.8	22.5	35606.2	0.151E+04
Web	12.0	230.0	135.8	27.6	374876.9	0.162E+04
BasePlate	150.0	16.0	258.8	24.0	621179.9	0.624E+04
Totals	...	...	...	107.5	1049724.4	0.122E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1049724 / 107.47$$

$$= 97.676 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 1195.332$$

$$= 243.2760 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 3.2831 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 530.0 - 263.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 312.356 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 75.9888 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 6.0793 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1195 + 55) \cdot 150.0) / (2 \cdot 500.0 \cdot 1767.415)}$$

$$= 5.642 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (260.0 + 14.0 + 10.0)$$

$$= 491.902 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nrings} - 1)$$

$$= 491.9024 / (3 - 1)$$

$$= 245.951 \text{ mm}$$



Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 245.9512 \cdot 150.0 / 2$$

$$= 184.463 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 1250.285 / (500.0 \cdot 150.0)$$

$$= 1.667 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 184.5 \cdot 1.67$$

$$= 307.509 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 16.56 + 14.757$$

$$= 31.317 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 307.5 / 31.3171$$

$$= 9.819 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	470.
Web	123.0	12.0	6.0	14.8	8854.2	234.
Totals	...	...	...	31.3	142990.2	704.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 12.0$$

$$= 138.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 142990.234 / 31.317$$

$$= 45.659 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{Saddle Width} - ytot)$$

$$= \max(45.659, 150.0 - 45.659)$$

$$= 104.341 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (703.5 / 31.317) }$$

$$= 47.396 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 530.0 - \cos(120.0 / 2) (260.0 + 14.0 + 10.0) - 16.0$$

$$= 372.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 2028094 / 2651.1) }$$

$$= 122.884$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 372.0 / 47.396$$

$$= 7.849$$



Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 24.9 / (2 \cdot 500.0) \cdot 245.951 \cdot 372.0 / 2$$

$$= 1.141 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (7.8487 < 122.8836) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.85)^2 / (2 \cdot 122.88^2)) 2651 /$$

$$(5/3 + 3 \cdot (7.85) / (8 \cdot 122.88) - (7.85^3) / (8 \cdot 122.88^3))$$

$$= 1565 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 9.82 / 1564.97 + (1.14 \cdot 104.341 / 7035148) / 1767.42$$

$$= 0.007$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	645.
Web	246.0	12.0	6.0	29.5	17708.5	218.
Totals	...	...	...	46.1	151844.5	863.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 151844.469 / 46.074$$

$$= 32.957 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(32.957, \text{abs}(150.0 - 32.957))$$

$$= 117.043 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 530.0 - \cos(284.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 230.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{863.0 / 46.074}$$

$$= 43.280 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 230.0 / 43.28$$

$$= 5.314$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 24.947 / (2 \cdot 500.0)$$

$$= 0.025 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.025 \cdot 245.951 \cdot 230.0$$

$$= 1.411 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1.411 / 73737.445$$

$$= 1.914 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (5.3142 < 122.8836) per AISC E2-1 [Sca]:

$$= (1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.31)^2 / (2 \cdot 122.88^2)) 2651 /$$

$$(5/3 + 3 \cdot (5.31) / (8 \cdot 122.88) - (5.31^3) / (8 \cdot 122.88^3))$$

$$= 1574 \text{ kgf/cm}^2$$



AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} &= Sc/Sca + (Mbase,c \cdot C1/I)/Sba \\ &= 14.29/1573.88 + (1.41 \cdot 117.043/863.048)/1767.42 \\ &= 0.010 \end{aligned}$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	20.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1407.13 kgf/cm ²
Area Available in a Single Bolt	BltArea	1.4584 cm ²
Saddle Load QO (Weight)	QO	980.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	6.2 kgf
Maximum Transverse Force	Ft	83.4 kgf
Maximum Longitudinal Force	Fl	24.9 kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$\begin{aligned} &= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 24.9 / (1.458 \cdot 2) \\ &= 8.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2. \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$\begin{aligned} &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 83.4 / (1.458 \cdot 2) \\ &= 28.6 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= Fl / (BoltShearAllowable $\cdot$ Nbolts)
= 24.95/(1052.29 $\cdot$ 2.0)
= 0.0119 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B $\cdot$ Ft + Sum of X Moments
= 530.0 $\cdot$ 83.41 + 0.0
= 44.21 kgf-m.

Eccentricity (e):
= Rmom / QO
= 44.21/980.68
= 45.08 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0119, 0.0]
= 0.0119 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

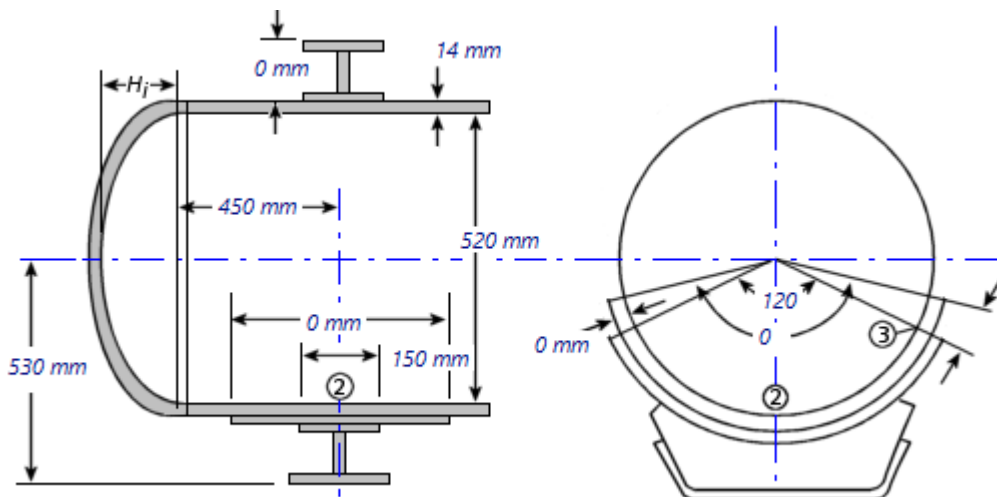
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{268.5875 \cdot 10.825}, 2 \cdot 450.0)$$

$$= 234.1165 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	268.59	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	320.88	cm
Distance from Saddle to Vessel tangent	a1 or a	450.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	530.00	mm
Coefficient of Friction	mu	0.50	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		2967.22	kgf
Pressure used in Analysis	P	-0.000	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-43.96	-2215.74
Long. Stress at Top of Midspan	43.96	1407.13
Long. Stress at Bottom of Midspan	43.96	1407.13
Long. Stress at Top of Saddles	58.84	1407.13
Long. Stress at Bottom of Saddles	-32.61	-2215.74
Long. Stress at Bottom of Saddles	32.61	1407.13
Tangential Shear in Shell	85.97	1125.71
Circ. Stress at Horn of Saddle	230.02	1758.92



Circ. Compressive Stress in Shell 8.90 1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(505.5/2 + 0) \cdot 530.0/491.9024$
 $= 817.0\ kgf$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(75.6, 0) \cdot 530.0/2148.0002$
 $= 18.7\ kgf$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(253.35, 0) \cdot 530.0/2148.0002$
 $= 62.5\ kgf$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(253/2 + 0) \cdot 530.0/491.9024$
 $= 409.5\ kgf$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 2150 + \max(19, 817, 63, 409)$
 $= 2967.2\ kgf$

Longitudinal Wind Force [FI]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1984.306(0.0 + \pi/4(0.548 \cdot 1.2)^2)$
 $= 741.341\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3022.18	kgf
Transverse Shear Load	252.76	kgf
Longitudinal Shear Load	253.35	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness)\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(520.0/2 + 14.0 + 10.0)\sin(120.0/2), 2 \cdot 268.587)$
 $= 491.902\ mm$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/(3L))]$
 $= -2967 \cdot 45.0 [1 - (1 - 45.0/320.88 + (26.859^2 - 0.0^2) / (2 \cdot 45.0 \cdot 320.88)) / (1 + (4 \cdot 0.0)/(3 \cdot 320.88))]$
 $= -153.9\ kgf \cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 [(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm)/(3L)) - 4a/L]$
 $= 2967 \cdot 321/4 [(1 + 2(27^2 - 0^2)/(321^2)) / (1 + (4 \cdot 0)/(3 \cdot 321)) - 4 \cdot 45/321]$
 $= 1078.4\ kgf \cdot m.$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 t)$
 $= -0.0 \cdot 268.587/(2 \cdot 10.825) - 1078.4/(\pi \cdot 268.6^2 \cdot 10.825)$
 $= -43.96\ kgf/cm^2$



Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0.0 \cdot 268.587 / (2 \cdot 10.825) + 1078.4 / (\pi \cdot 268.6^2 \cdot 10.825)$$

$$= 43.96 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.0 \cdot 268.587 / (2 \cdot 10.825) - 153.9 / (0.1066 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 58.84 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.0 \cdot 268.587 / (2 \cdot 10.825) + 153.9 / (0.1923 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= -32.61 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 2967(320.88 - 2 \cdot 45.0) / (320.88 + (4 \cdot 0.0 / 3))$$

$$= 2135.0 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2134.98 / (268.5875 \cdot 10.825)$$

$$= 85.97 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(268.587 \cdot 10.825)}$$

$$= 42.058 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 2967 \cdot 0.1 / (10.825(150.0 + 42.06 + 42.06))$$

$$= -8.90 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -2967 / (4 \cdot 10.825(150.0 + 42.058 + 42.058)) -$$

$$3 \cdot 0.0529 \cdot 2967 / (2 \cdot 10.825^2)$$

$$= -230.02 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(268.587 \cdot 10.825)}, 2 \cdot 45.0)$$

$$= 234.12 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	500.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2461.9	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(263.175 \cdot 10.825)} + 225.0) 10.825^3) / 12$$

$$= 3.259 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 3.259 + 33.37(97.676 - 5.413)^2$$



$$= 2843.888 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	308.3	10.8	5.4	33.4	18061.3	0.284E+04
Wearplate	225.0	10.0	15.8	22.5	35606.2	0.151E+04
Web	12.0	230.0	135.8	27.6	374876.9	0.162E+04
BasePlate	150.0	16.0	258.8	24.0	621179.9	0.624E+04
Totals	...	...	...	107.5	1049724.4	0.122E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1049724 / 107.47$$

$$= 97.676 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 2967.222$$

$$= 603.8940 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 8.1497 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1477.1587 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 530.0 - 263.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 312.356 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 188.6302 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 15.0908 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1641.2876 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(2967 + 55) \cdot 150.0) / (2 \cdot 500.0 \cdot 1641.288)}$$

$$= 9.103 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (260.0 + 14.0 + 10.0)$$

$$= 491.902 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 491.9024 / (3 - 1)$$

$$= 245.951 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 245.9512 \cdot 150.0 / 2$$

$$= 184.463 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3022.176 / (500.0 \cdot 150.0)$$

$$= 4.030 \text{ kgf/cm}^2$$



Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 184.5 \cdot 4.03$$

$$= 743.308 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 16.56 + 14.757$$

$$= 31.317 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 743.3/31.3171$$

$$= 23.735 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	470.
Web	123.0	12.0	6.0	14.8	8854.2	234.
Totals	...	...	...	31.3	142990.2	704.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 150.0 - 12.0$$

$$= 138.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 142990.234/31.317$$

$$= 45.659 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(45.659, 150.0 - 45.659)$$

$$= 104.341 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(703.5/31.317)}$$

$$= 47.396 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 530.0 - \cos(120.0/2) (260.0 + 14.0 + 10.0) - 16.0$$

$$= 372.000 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2028094/2461.9)}$$

$$= 127.518$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 372.0/47.396$$

$$= 7.849$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 253.4 / (2 \cdot 500.0) \cdot 245.951 \cdot 372.0/2$$

$$= 11.590 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ ($7.8487 < 127.5179$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.85)^2 / (2 \cdot 127.52^2)) 2462 / (5/3 + 3 \cdot (7.85) / (8 \cdot 127.52) - (7.85^3) / (8 \cdot 127.52^3))$$



= 1454 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= $Sc/Sca + (Rm \cdot C1 / I) / Sba$
 = $23.73/1454.25 + (11.59 \cdot 104.341/7035148) / 1641.29$
 = 0.027

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	645.
Web	246.0	12.0	6.0	29.5	17708.5	218.
Totals	...	...	...	46.1	151844.5	863.

Distance to Centroid from Datum [ytot]:

= AY / A
 = $151844.469/46.074$
 = 32.957 mm

Distance to Centroid [C1]:

= $\max(ytot, \text{abs}(Saddle\ Width - ytot))$
 = $\max(32.957, \text{abs}(150.0 - 32.957))$
 = 117.043 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = $530.0 - \cos(284.0/2) (16.0 + 0.0 + 0.0) - 0.0$
 = 230.000 mm

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$
 = $\sqrt{(863.0/46.074)}$
 = 43.280 mm

Slenderness ratio [KL/r]:

= KL/r
 = $1 \cdot 230.0/43.28$
 = 5.314

Unit Force [Force,u]:

= $FI / (2 \cdot Baseplate\ Length)$
 = $253.352/(2 \cdot 500.0)$
 = 0.253 kgf/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force $\cdot e \cdot L$
 = $0.253 \cdot 245.951 \cdot 230.0$
 = 14.332 kgf-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = $14.332/73737.445$
 = 19.436 kgf/cm²

Compressive Allowable, $KL/r < Cc$ (5.3142 < 127.5179) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (5.31)^2 / (2 \cdot 127.52^2)) 2462 /$
 $(5/3 + 3 \cdot (5.31) / (8 \cdot 127.52) - (5.31^3) / (8 \cdot 127.52^3))$
 = 1462 kgf/cm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (Mbase,c \cdot C1/I) / Sba$
 = $34.55/1462.17 + (14.33 \cdot 117.043/863.048) / 1641.29$
 = 0.035

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:



Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{ 268.5875 \cdot 10.825 }, 2 \cdot 450.0)$$

$$= 234.1165 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	268.59	mm
Shell Thickness used in this Case	t	14.000	mm
Stiffened Vessel Length per 4.15.6	L	320.88	cm
Distance from Saddle to Vessel tangent	a1 or a	450.00	mm
Saddle Width	b1 or b	150.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	13.86	cm
Shell Allowable Stress used in Calculation		1407.13	kgf/cm ²
Head Allowable Stress used in Calculation		1407.13	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	530.00	mm
Coefficient of Friction	mu	0.50	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		1535.03	kgf
Pressure used in Analysis	P	-0.000	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-19.79	-2215.74
Long. Stress at Top of Midspan		19.79	1407.13
Long. Stress at Bottom of Midspan		19.79	1407.13
Long. Stress at Top of Saddles		44.82	1407.13
Long. Stress at Bottom of Saddles		-24.84	-2215.74
Long. Stress at Bottom of Saddles		24.84	1407.13
Tangential Shear in Shell		42.05	1125.71
Circ. Stress at Horn of Saddle		118.99	1758.92
Circ. Compressive Stress in Shell		4.60	1407.13

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(505.5/2 + 0) \cdot 530.0/491.9024$$

$$= 817.0 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(75.6, 0) \cdot 530.0/2148.0002$$

$$= 18.7 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(253.35, 0) \cdot 530.0/2148.0002$$

$$= 62.5 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(253/2 + 0) \cdot 530.0/491.9024$$

$$= 409.5 \text{ kgf}$$



Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 718 + \max(19, 817, 63, 409)$$

$$= 1535.0 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.548 \cdot 1.2)^2)$$

$$= 741.341 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	1589.99	kgf
Transverse Shear Load	252.76	kgf
Longitudinal Shear Load	253.35	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(520.0/2 + 14.0 + 10.0)\sin(120.0/2), 2 \cdot 268.587)$$

$$= 491.902 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -1535 \cdot 45.0 [1 - (1 - 45.0/320.88 + (26.859^2 - 13.859^2)/(2 \cdot 45.0 \cdot 320.88))/(1 + (4 \cdot 13.86)/(3 \cdot 320.88))]$$

$$= -117.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 1535 \cdot 321/4((1 + 2(27^2 - 14^2)/(321^2))/(1 + (4 \cdot 14)/(3 \cdot 321)) - 4 \cdot 45/321)$$

$$= 485.6 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= -0.0 \cdot 268.587/(2 \cdot 10.825) - 485.6/(\pi \cdot 268.6^2 \cdot 10.825)$$

$$= -19.79 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= -0.0 \cdot 268.587/(2 \cdot 10.825) + 485.6/(\pi \cdot 268.6^2 \cdot 10.825)$$

$$= 19.79 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= -0.0 \cdot 268.587/(2 \cdot 10.825) - 117.2/(0.1066 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= 44.82 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0.0 \cdot 268.587/(2 \cdot 10.825) + 117.2/(0.1923 \cdot \pi \cdot 268.6^2 \cdot 10.825)$$

$$= -24.84 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 1535(320.88 - 2 \cdot 45.0)/(320.88 + (4 \cdot 13.86/3))$$

$$= 1044.3 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 1044.35/(268.5875 \cdot 10.825)$$

$$= 42.05 \text{ kgf/cm}^2$$



Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(268.587 \cdot 10.825)}$$

$$= 42.058 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 1535 \cdot 0.1 / (10.825(150.0 + 42.06 + 42.06))$$

$$= -4.60 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -1535 / (4 \cdot 10.825(150.0 + 42.058 + 42.058)) -$$

$$3 \cdot 0.0529 \cdot 1535 / (2 \cdot 10.825^2)$$

$$= -118.99 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(150.0 + 1.56 \cdot \sqrt{(268.587 \cdot 10.825)}, 2 \cdot 45.0)$$

$$= 234.12 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	500.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	12.0000	mm
Web Thickness	Webtk	12.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	2461.9	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(263.175 \cdot 10.825)} + 225.0)10.825^3) / 12$$

$$= 3.259 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 3.259 + 33.37(97.676 - 5.413)^2$$

$$= 2843.888 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	308.3	10.8	5.4	33.4	18061.3	0.284E+04
Wearplate	225.0	10.0	15.8	22.5	35606.2	0.151E+04
Web	12.0	230.0	135.8	27.6	374876.9	0.162E+04
BasePlate	150.0	16.0	258.8	24.0	621179.9	0.624E+04
Totals	...	...	...	107.5	1049724.4	0.122E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1049724 / 107.47$$

$$= 97.676 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$



= 0.2035

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 1535.035$$

$$= 312.4128 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 4.2161 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1477.1587 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 530.0 - 263.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 312.356 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 97.5841 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 7.8070 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1641.2876 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(1535 + 55)150.0) / (2 \cdot 500.0 \cdot 1641.288)}$$

$$= 6.603 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (260.0 + 14.0 + 10.0)$$

$$= 491.902 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 491.9024 / (3 - 1)$$

$$= 245.951 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 245.9512 \cdot 150.0/2$$

$$= 184.463 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 1589.988 / (500.0 \cdot 150.0)$$

$$= 2.120 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 184.5 \cdot 2.12$$

$$= 391.059 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 16.56 + 14.757$$

$$= 31.317 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 391.1/31.3171$$

$$= 12.487 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	470.
Web	123.0	12.0	6.0	14.8	8854.2	234.

Totals	...	...	...	31.3	142990.2	704.
--------	-----	-----	-----	------	----------	------

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 150.0 - 12.0
 = 138.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 142990.234/31.317
 = 45.659 mm

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(45.659, 150.0 - 45.659)
 = 104.341 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{703.5/31.317}$
 = 47.396 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 530.0 - cos(120.0/2)(260.0 + 14.0 + 10.0) - 16.0
 = 372.000 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2028094/2461.9}$
 = 127.518

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 372.0/47.396
 = 7.849

Bending Moment [Rm]:
 = $Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$
 = 253.4/(2 • 500.0) • 245.951 • 372.0/2
 = 11.590 kgf-m.

Compressive Allowable, $KL/r < Cc$ (7.8487 < 127.5179) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (7.85)^2 / (2 \cdot 127.52^2)) 2462 / (5/3 + 3 \cdot (7.85) / (8 \cdot 127.52) - (7.85^3) / (8 \cdot 127.52^3))$
 = 1454 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = $Sc/Sca + (Rm \cdot C1 / I) / Sba$
 = 12.49/1454.25 + (11.59 • 104.341/7035148)/1641.29
 = 0.019

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	12.0	138.0	81.0	16.6	134136.0	645.
Web	246.0	12.0	6.0	29.5	17708.5	218.
Totals	...	...	...	46.1	151844.5	863.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 151844.469/46.074
 = 32.957 mm

Distance to Centroid [C1]:
 = max(ytot, abs(Saddle Width - ytot))
 = max(32.957, abs(150.0 - 32.957))



$$= 117.043 \text{ mm}$$

Length of Inner Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk} \\ &= 530.0 - \cos(284.0/2) (16.0 + 0.0 + 0.0) - 0.0 \\ &= 230.000 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(863.0/46.074)} \\ &= 43.280 \text{ mm} \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL/r \\ &= 1 \cdot 230.0/43.28 \\ &= 5.314 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned} &= FI / (2 \cdot \text{Baseplate Length}) \\ &= 253.352 / (2 \cdot 500.0) \\ &= 0.253 \text{ kgf/mm} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} &= \text{Unit Force} \cdot e \cdot L \\ &= 0.253 \cdot 245.951 \cdot 230.0 \\ &= 14.332 \text{ kgf-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned} &= \text{Bending Moment} / \text{Section Modulus} \\ &= 14.332/73737.445 \\ &= 19.436 \text{ kgf/cm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($5.3142 < 127.5179$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (5.31)^2 / (2 \cdot 127.52^2)) 2462 / \\ &\quad (5/3 + 3 \cdot (5.31) / (8 \cdot 127.52) - (5.31^3) / (8 \cdot 127.52^3)) \\ &= 1462 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba} \\ &= 18.18/1462.17 + (14.33 \cdot 117.043/863.048)/1641.29 \\ &= 0.024 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: TS IN NOZZLE

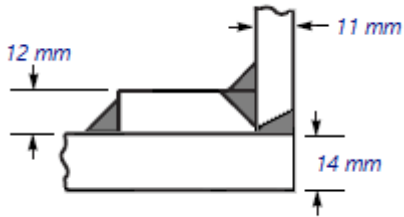
From: 20

Pressure for Reinforcement Calculations	P	36.287	kgf/cm ²
Temperature for Internal Pressure	Temp	66	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	520.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		30.00	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.9728	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	318.2750	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS IN NOZZLE

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $36.29 \cdot 263.175 / (1407 \cdot 1.0 - 0.6 \cdot 36.29)$
= 6.8934 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $36.29 \cdot 76.34 / (1203 \cdot 1.0 - 0.6 \cdot 36.29)$
= 2.3447 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588 mm
Parallel to Vessel Wall, opening length	d	152.6794 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.0625 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203.2 / 1407.1)$
= 0.855

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407.1 / 1407.1)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855

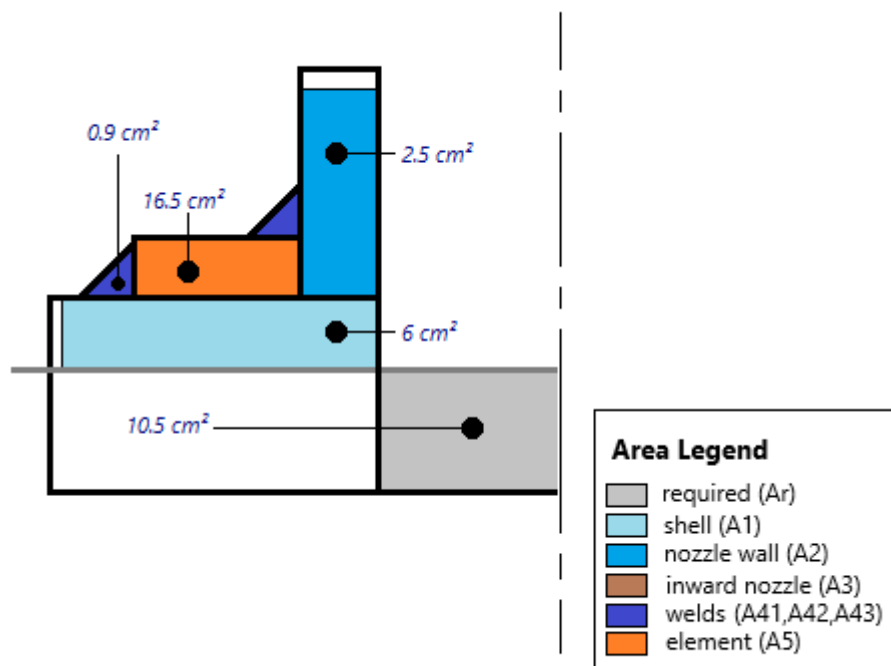


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10.525	NA	NA
Area in Shell	A1	6.003	NA	NA
Area in Nozzle Wall	A2	2.524	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	16.450	NA	NA
TOTAL AREA AVAILABLE	Atot	25.832	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 177.8012 12.0000 mm
 Based on given Pad Diameter: 318.2750 0.8339 mm
 Based on Shell or Nozzle Thickness: 178.6930 10.9728 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152.6794 \cdot 6.8934 \cdot 1.0 + 2 \cdot 7.7978 \cdot 6.8934 \cdot 1.0 \cdot (1 - 1.0))$
 $= 10.525 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.679(1.0 \cdot 10.825 - 1.0 \cdot 6.893) - 2 \cdot 7.798$
 $(1.0 \cdot 10.825 - 1.0 \cdot 6.893) \cdot (1 - 1.0)$
 $= 6.003 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 27.06) \cdot (7.8 - 2.34) \cdot 0.8551$



$$= 2.524 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 0.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (305.3588 - 168.275) \cdot 12.0 \cdot 1.0$
 $= 16.450 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$ta = 5.5197 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.6750 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 10.0684 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 10.0684 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.6750 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 9.3950 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.395, \max(10.0684, 4.675)]$
 $= 9.3950 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(5.5197, 9.395)$
 $= 9.3950 \text{ mm}$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 158.9,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2849.1 kgf/cm ²	Passed
Occasional	: 169.0,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 46.4,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.601$, $t_r = 2.345$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.365$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.601$, $t_r = 2.345$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.365$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.601$, $t_r = 2.345$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.365$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C



Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 3.175 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.637$, Temp. Reduction = 20 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 2.345, c = 3.175 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.365$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $36.29/104.01 = 0.349$

Weld Size Calculations, Description: TS IN NOZZLE

Intermediate Calc. for nozzle/shell Welds	Tmin	7.7978 mm
Intermediate Calc. for pad/shell Welds	TminPad	10.8250 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4585 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$5.4125 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (A_r - A_1) \cdot S_v$
 $= (10.5247 - 6.0028) \cdot 1407$
 $= 6362.98 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot S_v$
 $= (2.5237 + 16.4501 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 27901.82 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 15584. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 318.275 \cdot 10.0 \cdot 0.49 \cdot 1407$
 $= 34471. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 168.275 \cdot 12.0 \cdot 0.74 \cdot 1407$



= 33028. kgf

Shear, Nozzle Groove Weld [Sngw]:

$$= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng$$

$$= (3.14/2) \cdot (152.679+10.973) \cdot (10.973-3.175) \cdot 0.6 \cdot 1407$$

$$= 16924. \text{ kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNGW}) = (34471 + 16924) = 51395 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 51394 kgf, must exceed W = 6362 kgf or W1 = 27901 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 56 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 13.9901 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.9901 mm

Input Echo, WRC297 Item 1, Description: TS IN NOZZLE :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	520.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	10.9728	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	318.275	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	36.29	kgf/cm ²
Radial Load	P	-407.91	kgf
Circumferential Shear	Vc	611.86	kgf
Longitudinal Shear	Vl	611.86	kgf
Circumferential Moment	Mc	0.28	kgf-m.
Longitudinal Moment	Ml	0.37	kgf-m.
Torsional Moment	Mt	0.46	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{274.587(14.0 - 3.175)}$$

$$= 348.190 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	549.175 mm
Nozzle Outside Diameter	(d) =	168.275 mm
Vessel Thickness used	(T) =	22.825 mm
Nozzle Thickness used	(t) =	7.798 mm
T / t	=	2.927
d / t	=	21.580

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.503$$

Nr/P	=	0.082
Mr/P	=	0.084
M0/P	=	0.082
N0/P	=	0.200
MrD/Mc	=	0.170
NrDL/Mc	=	0.129
M0d/Mc	=	0.183
N0DL/Mc	=	0.148
MrD/M1	=	0.126
NrDL/M1	=	0.097
M0D/M1	=	0.137
N0DL/M1	=	0.285

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	38	-38	38	-38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	400	437	400	437
Outplane Stress Summary	454	414	455	414

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	200	200	200	200
Inplane Stress Summary	245	167	246	167

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10



Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	10	10	-10	-10
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	454	415	455	414

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	38	-38	38	-38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	200	200	200	200
Outplane Stress Summary	254	178	254	177

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	400	437	400	437
Inplane Stress Summary	446	404	446	404

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-10	-10	10	10

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	446	405	447	404

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	405	449	405	449
Hoop Stress Summary	421	464	421	464

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	171	-171	171	-171
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	202	202	202	202
Axial Stress Summary	382	43	385	40

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	29	29	-29	-29

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	437	466	438	466

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	405	449	405	449
Hoop Stress Summary	421	464	421	464

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	171	-171	171	-171
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	202	202	202	202
Axial Stress Summary	382	43	385	40

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-29	-29	29	29

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	437	466	438	466

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(268.587(14.0 - 3.175))}$
 $= 346.214 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	537.175 mm
Nozzle Outside Diameter	(d) =	318.275 mm
Vessel Thickness used	(T) =	10.825 mm
Nozzle Thickness used	(t) =	82.798 mm
T / t	=	0.131
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.174$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.080
M0/P	=	0.024
N0/P	=	0.039
MrD/Mc	=	0.251
NrDL/Mc	=	0.098
M0d/Mc	=	0.075
N0DL/Mc	=	0.067
MrD/Ml	=	0.061
NrDL/Ml	=	0.053
M0D/Ml	=	0.019
N0DL/Ml	=	0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	864	900	864	900
Outplane Stress Summary	927	865	927	865

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	166	-166	166	-166
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	432	432	432	432
Inplane Stress Summary	615	282	616	282

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	11	11	-11	-11

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	927	865	927	865

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	432	432	432	432
Outplane Stress Summary	494	396	495	396

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	166	-166	166	-166
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	864	900	864	900
Inplane Stress Summary	1047	752	1049	750

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-11	-11	11	11
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-11	-11	11	11

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 1047 752 1049 750

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	400	437	400	437	400	437	400	437
Circ. Pl (SUS)	15	15	15	15	6	6	6	6
Circ. Q (SUS)	38	-38	38	-38	39	-39	39	-39
Long. Pm (SUS)	200	200	200	200	200	200	200	200
Long. Pl (SUS)	6	6	6	6	15	15	15	15
Long. Q (SUS)	39	-39	39	-39	38	-38	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	400.0	437.0	400.0	437.0	400.0	437.0	400.0	437.0
Pm+Pl (SUS)	415.5	452.4	415.5	452.4	406.5	443.4	406.5	443.4
Pm+Pl+Q (Total)	453.5	414.4	453.5	414.4	445.5	404.4	445.5	404.4

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	437.00	1407.13	Passed
Pm+Pl (SUS)	452.41	2110.70	Passed
Pm+Pl+Q (TOTAL)	453.48	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	864	900	864	900	864	900	864	900
Circ. Pl (SUS)	13	13	13	13	17	17	17	17
Circ. Q (SUS)	49	-49	49	-49	165	-165	167	-167
Long. Pm (SUS)	432	432	432	432	432	432	432	432
Long. Pl (SUS)	17	17	17	17	13	13	13	13
Long. Q (SUS)	166	-166	166	-166	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	864.0	900.0	864.0	900.0	864.0	900.0	864.0	900.0
Pm+Pl (SUS)	877.3	913.3	877.3	913.3	881.3	917.3	881.3	917.3
Pm+Pl+Q (Total)	926	864	926	864	1046	752	1048	750

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	900.00	1407.13	Passed
Pm+Pl (SUS)	917.26	2110.70	Passed
Pm+Pl+Q (TOTAL)	1048.22	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	405	449	405	449	405	449	405	449
Circ. Pl (SUS)	15	15	15	15	15	15	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	202	202	202	202	202	202	202	202
Long. Pl (SUS)	10	10	10	10	10	10	10	10
Long. Q (SUS)	170	-170	172	-172	170	-170	172	-172
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	405.0	449.0	405.0	449.0	405.0	449.0	405.0	449.0
Pm+Pl (SUS)	424.0	467.3	424.0	467.3	424.0	467.3	424.0	467.3
Pm+Pl+Q (Total)	435.7	466.0	436.1	466.0	435.7	466.0	436.1	466.0

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	449.00	1203.20	Passed
Pm+Pl (SUS)	467.29	1804.80	Passed
Pm+Pl+Q (TOTAL)	465.98	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: TS OUT NOZZLE

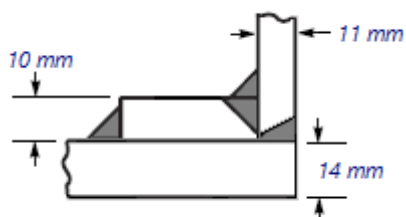
From: 20

Pressure for Reinforcement Calculations	P	36.250	kgf/cm ²
Temperature for Internal Pressure	Temp	66	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	520.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		30.00	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	318.2750	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: TS OUT NOZZLE

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.25 \cdot 263.175 / (1407 \cdot 1.0 - 0.6 \cdot 36.25)$
 $= 6.8862 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 36.25 \cdot 76.34 / (1203 \cdot 1.0 - 0.6 \cdot 36.25)$
 $= 2.3423 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588 mm
Parallel to Vessel Wall, opening length	d	152.6794 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.0625 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

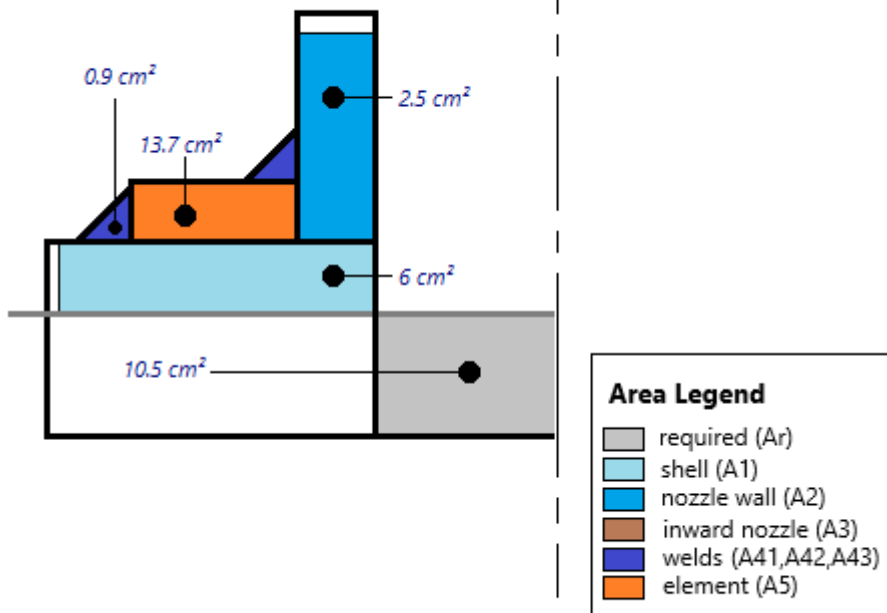


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10.514	NA	NA
Area in Shell	A1	6.014	NA	NA
Area in Nozzle Wall	A2	2.525	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	13.708	NA	NA
TOTAL AREA AVAILABLE	Atot	23.102	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 179.4778 10.0000 mm
 Based on given Pad Diameter: 318.2750 0.8172 mm
 Based on Shell or Nozzle Thickness: 178.4846 10.9728 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152.6794 \cdot 6.8862 \cdot 1.0 + 2 \cdot 7.7978 \cdot 6.8862 \cdot 1.0 \cdot (1 - 1.0))$
 $= 10.514 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.679(1.0 \cdot 10.825 - 1.0 \cdot 6.886) - 2 \cdot 7.798$
 $(1.0 \cdot 10.825 - 1.0 \cdot 6.886) \cdot (1 - 1.0)$
 $= 6.014 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 27.06) \cdot (7.8 - 2.34) \cdot 0.8551$



$$= 2.525 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can / 0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 0.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (305.3588 - 168.275) \cdot 10.0 \cdot 1.0$
 $= 13.708 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 5.5173 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.6750 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 10.0612 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 10.0612 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.6750 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 9.3950 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.395, \max(10.0612, 4.675)]$
 $= 9.3950 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(5.5173, 9.395)$
 $= 9.3950 \text{ mm}$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 158.7,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2849.3 kgf/cm ²	Passed
Occasional	: 168.8,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 46.4,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.601$, $t_r = 2.342$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.364$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.601$, $t_r = 2.342$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.364$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.601$, $t_r = 2.342$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.364$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C



Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 10.0, c = 3.175 mm, $E^* = 1.0$
Thickness Ratio = $tr * E^* / (tg - c) = 0.636$, Temp. Reduction = 20 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 2.342, c = 3.175 mm, $E^* = 1.0$
Thickness Ratio = $tr * E^* / (tg - c) = 0.364$, Temp. Reduction = 67 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $36.25/104.01 = 0.349$

Weld Size Calculations, Description: TS OUT NOZZLE

Intermediate Calc. for nozzle/shell Welds	Tmin	7.7978 mm
Intermediate Calc. for pad/shell Welds	TminPad	10.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4585 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$5.0000 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (A_r - A_1) * S_v$
 $= (10.5139 - 6.0137) * 1407$
 $= 6332.38$ kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 * fr_2) * S_v$
 $= (2.5248 + 13.7084 + 0.8551 - 0.0 * 0.86) * 1407$
 $= 24045.50$ kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) * D_{lo} * W_o * 0.49 * S_{nw}$
 $= (3.1416/2.0) * 168.275 * 10.0 * 0.49 * 1203$
 $= 15584.$ kgf

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) * D_P * W_P * 0.49 * S_{EW}$
 $= (3.1416/2.0) * 318.275 * 10.0 * 0.49 * 1407$
 $= 34471.$ kgf

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) * D_{lo} * W_{gpn} * 0.74 * S_{eg}$
 $= (3.1416/2) * 168.275 * 10.0 * 0.74 * 1407$
 $= 27524.$ kgf



Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNGW}) = (34471 + 0) = 34471 \text{ kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 34470 kgf, must exceed W = 6332 kgf or W1 = 24045 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 56 kgf/cm²

The Drop for this Nozzle is : 13.9901 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.9901 mm

Input Echo, WRC297 Item 1, Description: TS OUT NOZZLE :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	520.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	10.9728	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	318.275	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	36.25	kgf/cm ²
Radial Load	P	-407.91	kgf
Circumferential Shear	Vc	611.86	kgf
Longitudinal Shear	Vl	611.86	kgf
Circumferential Moment	Mc	0.28	kgf-m.
Longitudinal Moment	Ml	0.37	kgf-m.
Torsional Moment	Mt	0.46	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(273.587(14.0 - 3.175))}$$

$$= 347.862 \text{ mm}$$



WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 547.175 mm
 Nozzle Outside Diameter (d) = 168.275 mm
 Vessel Thickness used (T) = 20.825 mm
 Nozzle Thickness used (t) = 7.798 mm
 T / t = 2.671
 d / t = 21.580

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.576$$

Nr/P = 0.080
 Mr/P = 0.084
 M0/P = 0.073
 N0/P = 0.185
 MrD/Mc = 0.178
 NrDL/Mc = 0.129
 M0d/Mc = 0.170
 N0DL/Mc = 0.150
 MrD/Ml = 0.128
 NrDL/Ml = 0.093
 M0D/Ml = 0.123
 N0DL/Ml = 0.280

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	40	-40	40	-40
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	440	476	440	476
Outplane Stress Summary	498	453	499	453

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	220	220	220	220
Inplane Stress Summary	274	180	275	179

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	11	11	-11	-11

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 498 454 499 453

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	40	-40	40	-40
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	220	220	220	220
Outplane Stress Summary	278	197	278	196

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	440	476	440	476
Inplane Stress Summary	495	437	495	436

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-11	-11	11	11
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-11	-11	11	11

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	495	437	496	437

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	405	448	405	448
Hoop Stress Summary	422	465	422	466

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



Axial Membrane (P)	10	10	10	10
Axial Bending (P)	177	-177	177	-177
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	202	202	202	202

Axial Stress Summary	388	36	392	33
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	29	29	-29	-29

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	439	467	440	468

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	405	448	405	448
Hoop Stress Summary	422	466	422	466

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	177	-177	177	-177
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	202	202	202	202
Axial Stress Summary	388	36	392	33

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-29	-29	29	29

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside



Two * Max Shear Stress 439 468 440 468

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{268.587(14.0 - 3.175)}$$

$$= 346.214 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 537.175 mm
 Nozzle Outside Diameter (d) = 318.275 mm
 Vessel Thickness used (T) = 10.825 mm
 Nozzle Thickness used (t) = 82.798 mm
 T / t = 0.131
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 4.174$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.049
 Mr/P = 0.080
 M0/P = 0.024
 N0/P = 0.039
 MrD/Mc = 0.251
 NrDL/Mc = 0.098
 M0d/Mc = 0.075
 N0DL/Mc = 0.067
 MrD/Ml = 0.061
 NrDL/Ml = 0.053
 M0D/Ml = 0.019
 N0DL/Ml = 0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	863	899	863	899
Outplane Stress Summary	926	864	926	864

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	166	-166	166	-166
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	431	431	431	431
Inplane Stress Summary	615	282	615	281

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	11	11	-11	-11
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	926	864	926	864
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	431	431	431	431

Outplane Stress Summary	494	396	494	395
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	166	-166	166	-166
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	863	899	863	899

Inplane Stress Summary	1046	751	1048	749
------------------------	------	-----	------	-----

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-11	-11	11	11
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-11	-11	11	11
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1046	751	1048	749
------------------------	------	-----	------	-----

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	440	476	440	476	440	476	440	476
Circ. Pl (SUS)	17	17	17	17	7	7	7	7
Circ. Q (SUS)	40	-40	40	-40	47	-47	47	-47
Long. Pm (SUS)	220	220	220	220	220	220	220	220
Long. Pl (SUS)	7	7	7	7	17	17	17	17
Long. Q (SUS)	47	-47	47	-47	40	-40	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	440.0	476.0	440.0	476.0	440.0	476.0	440.0	476.0
Pm+Pl (SUS)	457.5	493.5	457.5	493.5	447.6	483.5	447.6	483.5
Pm+Pl+Q (Total)	497.5	453.4	497.5	453.4	494.6	436.5	494.6	436.5

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	476.00	1407.13	Passed
Pm+Pl (SUS)	493.45	2110.70	Passed
Pm+Pl+Q (TOTAL)	497.54	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	863	899	863	899	863	899	863	899
Circ. Pl (SUS)	13	13	13	13	17	17	17	17
Circ. Q (SUS)	49	-49	49	-49	165	-165	167	-167
Long. Pm (SUS)	431	431	431	431	431	431	431	431
Long. Pl (SUS)	17	17	17	17	13	13	13	13
Long. Q (SUS)	166	-166	166	-166	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	863.0	899.0	863.0	899.0	863.0	899.0	863.0	899.0
Pm+Pl (SUS)	876.3	912.3	876.3	912.3	880.3	916.3	880.3	916.3
Pm+Pl+Q (Total)	925	863	925	863	1045	751	1047	749

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	899.00	1407.13	Passed
Pm+Pl (SUS)	916.26	2110.70	Passed
Pm+Pl+Q (TOTAL)	1047.22	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:



Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	405	448	405	448	405	448	405	448
Circ. Pl (SUS)	17	17	17	17	17	17	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	202	202	202	202	202	202	202	202
Long. Pl (SUS)	10	10	10	10	10	10	10	10
Long. Q (SUS)	176	-176	178	-178	176	-176	178	-178
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	405.0	448.0	405.0	448.0	405.0	448.0	405.0	448.0
Pm+Pl (SUS)	425.9	468.3	425.9	468.3	425.9	468.3	425.9	468.3
Pm+Pl+Q (Total)	438.6	467.0	439.1	466.9	438.6	467.0	439.1	466.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	448.00	1203.20	Passed
Pm+Pl (SUS)	468.28	1804.80	Passed
Pm+Pl+Q (TOTAL)	466.95	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SH. IN NOZ-1

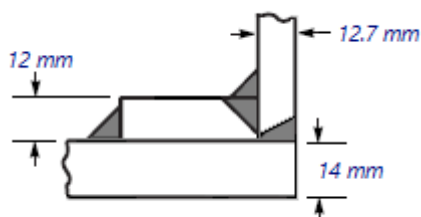
From: 50

Pressure for Reinforcement Calculations	P	47.200	kgf/cm ²
Temperature for Internal Pressure	Temp	88	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	520.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		171.32	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	369.0750	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH. IN NOZ-1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 7.625 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 47.2 \cdot 263.175 / (1407 \cdot 1.0 - 0.6 \cdot 47.2)$
 $= 9.0091 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 47.2 \cdot 100.01 / (1203 \cdot 1.0 - 0.6 \cdot 47.2)$
 $= 4.0179 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	400.0500	mm
Parallel to Vessel Wall, opening length	d	200.0250	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.0625	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

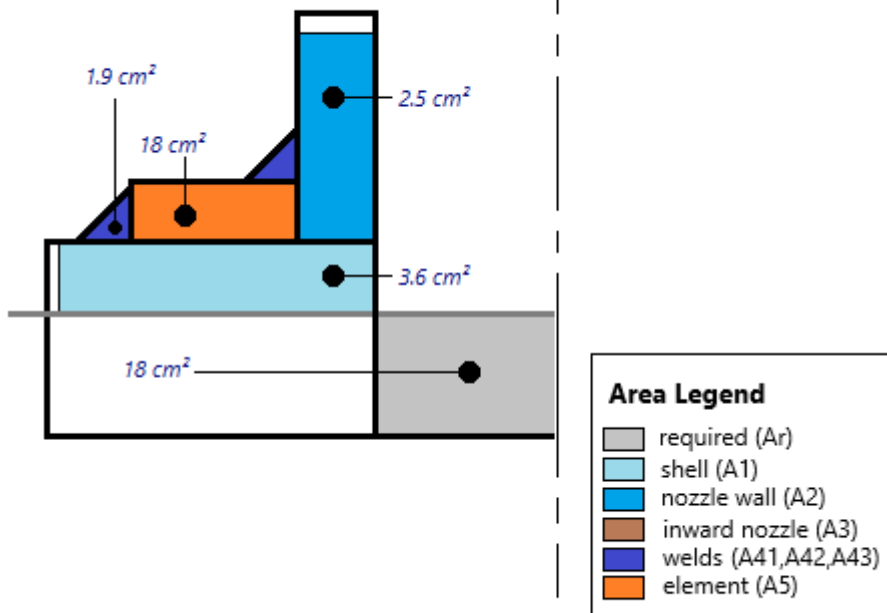


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18.020	NA	NA
Area in Shell	A1	3.632	NA	NA
Area in Nozzle Wall	A2	2.549	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.855	NA	NA
Area in Element	A5	18.000	NA	NA
TOTAL AREA AVAILABLE	Atot	26.036	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	302.2780	12.0000 mm
Based on given Pad Diameter:	369.0750	6.6562 mm
Based on Shell or Nozzle Thickness:	297.6920	12.7000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (200.025 \cdot 9.0091 \cdot 1.0 + 2 \cdot 9.525 \cdot 9.0091 \cdot 1.0 \cdot (1 - 1.0))$
 $= 18.020 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 200.025(1.0 \cdot 10.825 - 1.0 \cdot 9.0091) - 2 \cdot 9.525$
 $(1.0 \cdot 10.825 - 1.0 \cdot 9.0091) \cdot (1 - 1.0)$
 $= 3.632 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 27.06) \cdot (9.52 - 4.02) \cdot 0.8551$



$$= 2.549 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$
 $= 1.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (369.075 - 219.075) \cdot 12.0 \cdot 1.0$
 $= 18.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 7.1929 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.6750 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 12.1841 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 12.1841 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.6750 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 10.3350 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[10.335, \max(12.1841, 4.675)]$
 $= 10.3350 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(7.1929, 10.335)$
 $= 10.3350 \text{ mm}$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 228.0,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2780.0 kgf/cm ²	Passed
Occasional	: 236.5,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 38.7,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.113$, $t_r = 4.018$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 11.113$, $t_r = 4.018$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 11.113$, $t_r = 4.018$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $tr * E^* / (t_g - c) = 0.832$, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -32 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 11.113$, $tr = 4.018$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $tr * E^* / (t_g - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -37 °C
Gov. MDMT of the Reinforcement Pad : -32 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $47.20/104.01 = 0.454$

Weld Size Calculations, Description: SH. IN NOZ-1

Intermediate Calc. for nozzle/shell Welds T_{min} 9.5250 mm
Intermediate Calc. for pad/shell Welds T_{minPad} 10.8250 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$5.4125 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (A_r - A_1) * S_v$
 $= (18.0204 - 3.6323) * 1407$
 $= 20246.03 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - C_{an} / .707)^2 * f_{r2}) * S_v$
 $= (2.5487 + 18.0 + 1.8551 - 0.0 * 0.86) * 1407$
 $= 31525.10 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) * D_{lo} * W_o * 0.49 * S_{nw}$
 $= (3.1416/2.0) * 219.075 * 10.0 * 0.49 * 1203$
 $= 20288. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) * D_P * W_P * 0.49 * S_{EW}$
 $= (3.1416/2.0) * 369.075 * 10.0 * 0.49 * 1407$
 $= 39973. \text{ kgf}$



Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 219.075 \cdot 12.0 \cdot 0.74 \cdot 1407$
 $= 42999. \text{ kgf}$

Shear, Nozzle Groove Weld [Sngw]:
 $= (\pi/2) \cdot (D_{lr} + W_{gnva}) \cdot (W_{gnva} - C_{an}) \cdot 0.60 \cdot S_{ng}$
 $= (3.14/2) \cdot (200.025 + 12.0) \cdot (12.0 - 3.175) \cdot 0.6 \cdot 1407$
 $= 24815. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{NGW}) = (39973 + 24815) = 64788 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 64787 kgf, must exceed W = 20246 kgf or W1 = 31525 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 56 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 24.2002 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.2002 mm

Input Echo, WRC297 Item 1, Description: SH. IN NOZ-1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	520.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	193.675	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	369.075	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	47.20	kgf/cm ²
Radial Load	P	-550.67	kgf
Circumferential Shear	Vc	815.81	kgf
Longitudinal Shear	Vl	815.81	kgf
Circumferential Moment	Mc	0.49	kgf-m.
Longitudinal Moment	Ml	0.65	kgf-m.
Torsional Moment	Mt	0.82	kgf-m.



Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{274.587(14.0 - 3.175)}$
 $= 398.990 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 549.175 mm
 Nozzle Outside Diameter (d) = 219.075 mm
 Vessel Thickness used (T) = 22.825 mm
 Nozzle Thickness used (t) = 9.525 mm
 T / t = 2.396
 d / t = 23.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.957$$

Nr/P = 0.074
 Mr/P = 0.075
 M0/P = 0.057
 N0/P = 0.169
 MrD/Mc = 0.180
 NrDL/Mc = 0.122
 M0d/Mc = 0.150
 N0DL/Mc = 0.165
 MrD/Ml = 0.118
 NrDL/Ml = 0.077
 M0D/Ml = 0.098
 N0DL/Ml = 0.275

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	36	-36	36	-36
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	521	568	521	568
Outplane Stress Summary	575	550	576	550

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	260	260	260	260
Inplane Stress Summary	315	221	316	220

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	10	10	-10	-10

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	575	550	576	550

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	36	-36	36	-36
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	260	260	260	260
Outplane Stress Summary	314	242	315	241

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	521	568	521	568
Inplane Stress Summary	576	529	577	528

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-10	-10	10	10

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	576	529	577	528

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0



Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	567	624	567	624

Hoop Stress Summary	585	642	585	642
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	8	8	8	8
Axial Bending (P)	138	-138	138	-138
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	283	283	283	283

Axial Stress Summary	428	155	432	153
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	24	24	-24	-24
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	24	24	-24	-24
----------------------	----	----	-----	-----

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	589	643	589	643
------------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	567	624	567	624

Hoop Stress Summary	585	642	585	642
---------------------	-----	-----	-----	-----

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	8	8	8	8
Axial Bending (P)	138	-138	138	-138
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	283	283	283	283

Axial Stress Summary	428	156	432	152
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Nozzle Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-24	-24	24	24
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-24	-24	24	24

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	589	643	589	643

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(268.587(14.0 - 3.175))}$$

$$= 397.014 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	537.175 mm
Nozzle Outside Diameter (d)	=	369.075 mm
Vessel Thickness used (T)	=	10.825 mm
Nozzle Thickness used (t)	=	84.525 mm
T / t	=	0.128
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.840$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.070
M0/P	=	0.021
N0/P	=	0.035
MrD/Mc	=	0.244
NrDL/Mc	=	0.094
M0d/Mc	=	0.073
N0DL/Mc	=	0.064
MrD/Ml	=	0.054
NrDL/Ml	=	0.045
M0D/Ml	=	0.017
N0DL/Ml	=	0.033

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	58	-58	58	-58
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1124	1171	1124	1171
Outplane Stress Summary	1199	1129	1199	1129

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	20	20	20	20



Inplane Bending (P)	196	-196	196	-196
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	562	562	562	562

Inplane Stress Summary	778	387	779	386
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	12	12	-12	-12
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	1199	1129	1200	1129
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	58	-58	58	-58
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	562	562	562	562

Outplane Stress Summary	636	520	638	519
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	20	20	20	20
Inplane Bending (P)	196	-196	196	-196
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1124	1171	1124	1171

Inplane Stress Summary	1339	997	1343	994
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-12	-12	12	12
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	1339	998	1343	995

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	521	568	521	568	521	568	521	568
Circ. Pl (SUS)	17	17	17	17	7	7	7	7
Circ. Q (SUS)	36	-36	36	-36	47	-47	47	-47
Long. Pm (SUS)	260	260	260	260	260	260	260	260
Long. Pl (SUS)	7	7	7	7	17	17	17	17
Long. Q (SUS)	47	-47	47	-47	36	-36	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	521.0	568.0	521.0	568.0	521.0	568.0	521.0	568.0
Pm+Pl (SUS)	538.4	585.3	538.4	585.3	528.4	575.3	528.4	575.3
Pm+Pl+Q (Total)	574.4	549.3	574.4	549.3	575.4	528.3	575.4	528.3

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	568.00	1407.13	Passed
Pm+Pl (SUS)	585.31	2110.70	Passed
Pm+Pl+Q (TOTAL)	575.38	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1124	1171	1124	1171	1124	1171	1124	1171
Circ. Pl (SUS)	16	16	16	16	20	20	20	20
Circ. Q (SUS)	58	-58	58	-58	195	-195	197	-197
Long. Pm (SUS)	562	562	562	562	562	562	562	562
Long. Pl (SUS)	20	20	20	20	16	16	16	16
Long. Q (SUS)	196	-196	196	-196	58	-58	58	-58
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1124	1171	1124	1171	1124	1171	1124	1171
Pm+Pl (SUS)	1140	1187	1140	1187	1144	1191	1144	1191
Pm+Pl+Q (Total)	1198	1129	1198	1129	1339	996	1341	994

Vessel Stress Summation Comparison (kgf/cm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1171.00	1407.13	Passed
Pm+Pl (SUS)	1191.23	2110.70	Passed
Pm+Pl+Q (TOTAL)	1341.20	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	567	624	567	624	567	624	567	624
Circ. Pl (SUS)	17	17	17	17	17	17	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	283	283	283	283	283	283	283	283
Long. Pl (SUS)	8	8	8	8	8	8	8	8
Long. Q (SUS)	137	-137	139	-139	137	-137	139	-139
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	24	24	-24	-24	-24	-24	24	24
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	567.0	624.0	567.0	624.0	567.0	624.0	567.0	624.0
Pm+Pl (SUS)	586.0	642.6	586.0	642.6	586.0	642.6	586.0	642.6
Pm+Pl+Q (Total)	587.6	642.2	587.7	642.2	587.6	642.2	587.7	642.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	624.00	1203.20	Passed
Pm+Pl (SUS)	642.64	1804.80	Passed
Pm+Pl+Q (TOTAL)	642.18	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SH IN NOZ-2

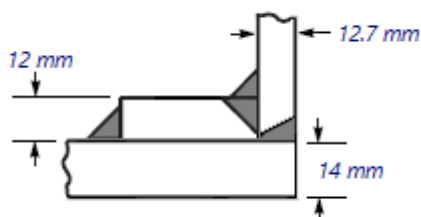
From: 50

Pressure for Reinforcement Calculations	P	47.200	kgf/cm ²
Temperature for Internal Pressure	Temp	88	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	520.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		306.12	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	369.0750	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH IN NOZ-2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 7.625 in.
 Actual Thickness Used in Calculation 0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 47.2 \cdot 263.175 / (1407 \cdot 1.0 - 0.6 \cdot 47.2)$
 $= 9.0091 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 47.2 \cdot 100.01 / (1203 \cdot 1.0 - 0.6 \cdot 47.2)$
 $= 4.0179 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	400.0500	mm
Parallel to Vessel Wall, opening length	d	200.0250	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.0625	mm

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

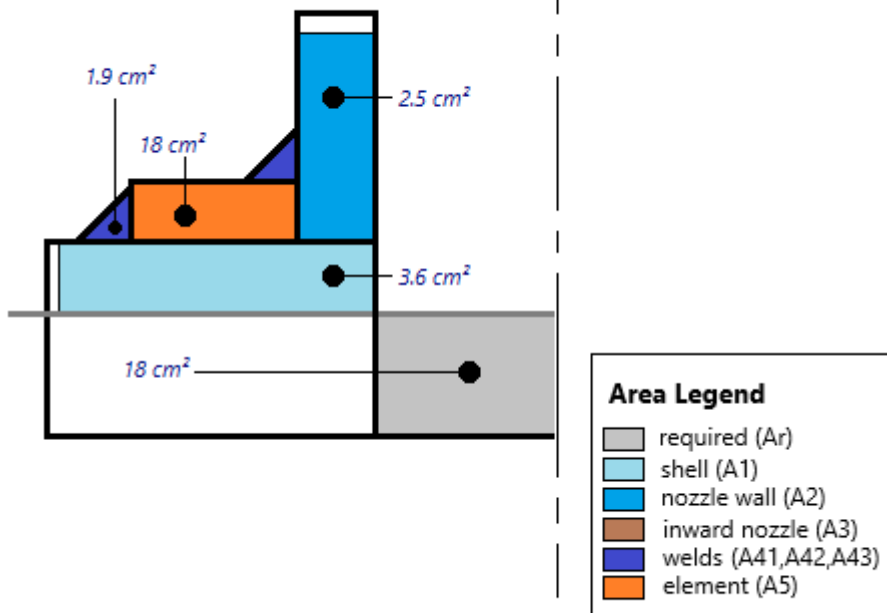


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	18.020	NA	NA
Area in Shell	A1	3.632	NA	NA
Area in Nozzle Wall	A2	2.549	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.855	NA	NA
Area in Element	A5	18.000	NA	NA
TOTAL AREA AVAILABLE	Atot	26.036	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 302.2780 12.0000 mm
 Based on given Pad Diameter: 369.0750 6.6562 mm
 Based on Shell or Nozzle Thickness: 297.6920 12.7000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (200.025 \cdot 9.0091 \cdot 1.0 + 2 \cdot 9.525 \cdot 9.0091 \cdot 1.0 \cdot (1 - 1.0))$
 $= 18.020 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 200.025(1.0 \cdot 10.825 - 1.0 \cdot 9.009) - 2 \cdot 9.525$
 $(1.0 \cdot 10.825 - 1.0 \cdot 9.009) \cdot (1 - 1.0)$
 $= 3.632 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 27.06) \cdot (9.52 - 4.02) \cdot 0.8551$



$$= 2.549 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 10.0^2 \cdot 1.0$
 $= 1.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (369.075 - 219.075) \cdot 12.0 \cdot 1.0$
 $= 18.000 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 7.1929 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.6750 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 12.1841 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 12.1841 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.6750 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 10.3350 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[10.335, \max(12.1841, 4.675)]$
 $= 10.3350 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(7.1929, 10.335)$
 $= 10.3350 \text{ mm}$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 12.700 = 11.113 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 228.0,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2780.0 kgf/cm ²	Passed
Occasional	: 236.5,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 38.7,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 11.113$, $t_r = 4.018$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 11.113$, $t_r = 4.018$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 11.113$, $t_r = 4.018$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C



Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.832$, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -23 °C
Min Metal Temp. at Required thickness (UCS 66.1) -32 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 11.113$, $t_r = 4.018$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $t_r * E^* / (t_g - c) = 0.506$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -37 °C
Gov. MDMT of the Reinforcement Pad : -32 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $47.20/104.01 = 0.454$

Weld Size Calculations, Description: SH IN NOZ-2

Intermediate Calc. for nozzle/shell Welds T_{min} 9.5250 mm
Intermediate Calc. for pad/shell Welds T_{minPad} 10.8250 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$5.4125 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (A_r - A_1) * S_v$
 $= (18.0204 - 3.6323) * 1407$
 $= 20246.03 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - C_{an} / .707)^2 * f_{r2}) * S_v$
 $= (2.5487 + 18.0 + 1.8551 - 0.0 * 0.86) * 1407$
 $= 31525.10 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) * D_{lo} * W_o * 0.49 * S_{nw}$
 $= (3.1416/2.0) * 219.075 * 10.0 * 0.49 * 1203$
 $= 20288. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) * D_P * W_P * 0.49 * S_{EW}$
 $= (3.1416/2.0) * 369.075 * 10.0 * 0.49 * 1407$
 $= 39973. \text{ kgf}$



Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 219.075 \cdot 12.0 \cdot 0.74 \cdot 1407$
 $= 42999. \text{ kgf}$

Shear, Nozzle Groove Weld [Sngw]:
 $= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng$
 $= (3.14/2) \cdot (200.025+12.0) \cdot (12.0-3.175) \cdot 0.6 \cdot 1407$
 $= 24815. \text{ kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNGW) = (39973 + 24815) = 64788 \text{ kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 64787 kgf, must exceed W = 20246 kgf or W1 = 31525 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 56 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 24.2002 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 238.2002 mm

Input Echo, WRC297 Item 1, Description: SH IN NOZ-2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	520.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	193.675	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	369.075	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	47.20	kgf/cm ²
Radial Load	P	-550.67	kgf
Circumferential Shear	Vc	815.81	kgf
Longitudinal Shear	Vl	815.81	kgf
Circumferential Moment	Mc	0.49	kgf-m.
Longitudinal Moment	Ml	0.65	kgf-m.
Torsional Moment	Mt	0.82	kgf-m.



Include Axial Pressure Thrust No
 Include Pressure Stress Indices per Div. 2 No
 Local Loads applied at end of Nozzle/Attachment No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(274.587(14.0 - 3.175))}$
 $= 398.990 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 549.175 mm
 Nozzle Outside Diameter (d) = 219.075 mm
 Vessel Thickness used (T) = 22.825 mm
 Nozzle Thickness used (t) = 9.525 mm
 T / t = 2.396
 d / t = 23.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.957$$

Nr/P = 0.074
 Mr/P = 0.075
 M0/P = 0.057
 N0/P = 0.169
 MrD/Mc = 0.180
 NrDL/Mc = 0.122
 M0d/Mc = 0.150
 N0DL/Mc = 0.165
 MrD/Ml = 0.118
 NrDL/Ml = 0.077
 M0D/Ml = 0.098
 N0DL/Ml = 0.275

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	36	-36	36	-36
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	521	568	521	568
Outplane Stress Summary	575	550	576	550

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	260	260	260	260
Inplane Stress Summary	315	221	316	220

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
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longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	10	10	-10	-10
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	10	10	-10	-10

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	575	550	576	550

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	17	17	17	17
Outplane Bending (P)	36	-36	36	-36
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	260	260	260	260
Outplane Stress Summary	314	242	315	241

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	521	568	521	568
Inplane Stress Summary	576	529	577	528

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-10	-10	10	10

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	576	529	577	528

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0



Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	567	624	567	624

Hoop Stress Summary	585	642	585	642
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to	Top	Top	Bottom	Bottom
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	8	8	8	8
Axial Bending (P)	138	-138	138	-138
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	283	283	283	283

Axial Stress Summary	428	155	432	153
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	24	24	-24	-24
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	24	24	-24	-24
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	589	643	589	643
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	17	17	17	17
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	567	624	567	624

Hoop Stress Summary	585	642	585	642
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	8	8	8	8
Axial Bending (P)	138	-138	138	-138
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	283	283	283	283

Axial Stress Summary	428	156	432	152
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Nozzle Stresses (kgf/cm²):



CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-24	-24	24	24
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-24	-24	24	24

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	589	643	589	643

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(268.587(14.0 - 3.175))}$
 $= 397.014 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	537.175 mm
Nozzle Outside Diameter (d)	=	369.075 mm
Vessel Thickness used (T)	=	10.825 mm
Nozzle Thickness used (t)	=	84.525 mm
T / t	=	0.128
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.840$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.070
M0/P	=	0.021
N0/P	=	0.035
MrD/Mc	=	0.244
NrDL/Mc	=	0.094
M0d/Mc	=	0.073
N0DL/Mc	=	0.064
MrD/Ml	=	0.054
NrDL/Ml	=	0.045
M0D/Ml	=	0.017
N0DL/Ml	=	0.033

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	58	-58	58	-58
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1124	1171	1124	1171
Outplane Stress Summary	1199	1129	1199	1129

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	20	20	20	20



Inplane Bending (P)	196	-196	196	-196
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	562	562	562	562

Inplane Stress Summary	778	387	779	386
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	12	12	-12	-12
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	12	12	-12	-12

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1199	1129	1200	1129

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	58	-58	58	-58
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	562	562	562	562
Outplane Stress Summary	636	520	638	519

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	20	20	20	20
Inplane Bending (P)	196	-196	196	-196
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1124	1171	1124	1171
Inplane Stress Summary	1339	997	1343	994

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-12	-12	12	12
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-12	-12	12	12

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl



(Stress Intensities	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	1339	998	1343	995

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	521	568	521	568	521	568	521	568
Circ. Pl (SUS)	17	17	17	17	7	7	7	7
Circ. Q (SUS)	36	-36	36	-36	47	-47	47	-47
Long. Pm (SUS)	260	260	260	260	260	260	260	260
Long. Pl (SUS)	7	7	7	7	17	17	17	17
Long. Q (SUS)	47	-47	47	-47	36	-36	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	521.0	568.0	521.0	568.0	521.0	568.0	521.0	568.0
Pm+Pl (SUS)	538.4	585.3	538.4	585.3	528.4	575.3	528.4	575.3
Pm+Pl+Q (Total)	574.4	549.3	574.4	549.3	575.4	528.3	575.4	528.3

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	568.00	1407.13	Passed
Pm+Pl (SUS)	585.31	2110.70	Passed
Pm+Pl+Q (TOTAL)	575.38	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1124	1171	1124	1171	1124	1171	1124	1171
Circ. Pl (SUS)	16	16	16	16	20	20	20	20
Circ. Q (SUS)	58	-58	58	-58	195	-195	197	-197
Long. Pm (SUS)	562	562	562	562	562	562	562	562
Long. Pl (SUS)	20	20	20	20	16	16	16	16
Long. Q (SUS)	196	-196	196	-196	58	-58	58	-58
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	12	12	-12	-12	-12	-12	12	12
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1124	1171	1124	1171	1124	1171	1124	1171
Pm+Pl (SUS)	1140	1187	1140	1187	1144	1191	1144	1191
Pm+Pl+Q (Total)	1198	1129	1198	1129	1339	996	1341	994

Vessel Stress Summation Comparison (kgf/cm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1171.00	1407.13	Passed
Pm+Pl (SUS)	1191.23	2110.70	Passed
Pm+Pl+Q (TOTAL)	1341.20	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	567	624	567	624	567	624	567	624
Circ. Pl (SUS)	17	17	17	17	17	17	17	17
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	283	283	283	283	283	283	283	283
Long. Pl (SUS)	8	8	8	8	8	8	8	8
Long. Q (SUS)	137	-137	139	-139	137	-137	139	-139
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	24	24	-24	-24	-24	-24	24	24
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	567.0	624.0	567.0	624.0	567.0	624.0	567.0	624.0
Pm+Pl (SUS)	586.0	642.6	586.0	642.6	586.0	642.6	586.0	642.6
Pm+Pl+Q (Total)	587.6	642.2	587.7	642.2	587.6	642.2	587.7	642.2

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	624.00	1203.20	Passed
Pm+Pl (SUS)	642.64	1804.80	Passed
Pm+Pl+Q (TOTAL)	642.18	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SH OUT NOZ-1

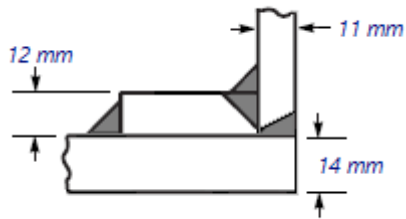
From: 50

Pressure for Reinforcement Calculations	P	47.201	kgf/cm ²
Temperature for Internal Pressure	Temp	88	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	520.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		171.32	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.9728	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	318.2750	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH OUT NOZ-1

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 47.2 \cdot 263.175 / (1407 \cdot 1.0 - 0.6 \cdot 47.2)$
 $= 9.0093 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 47.2 \cdot 76.34 / (1203 \cdot 1.0 - 0.6 \cdot 47.2)$
 $= 3.0670 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588 mm
Parallel to Vessel Wall, opening length	d	152.6794 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.0625 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 1203.2 / 1407.1)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 1407.1 / 1407.1)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

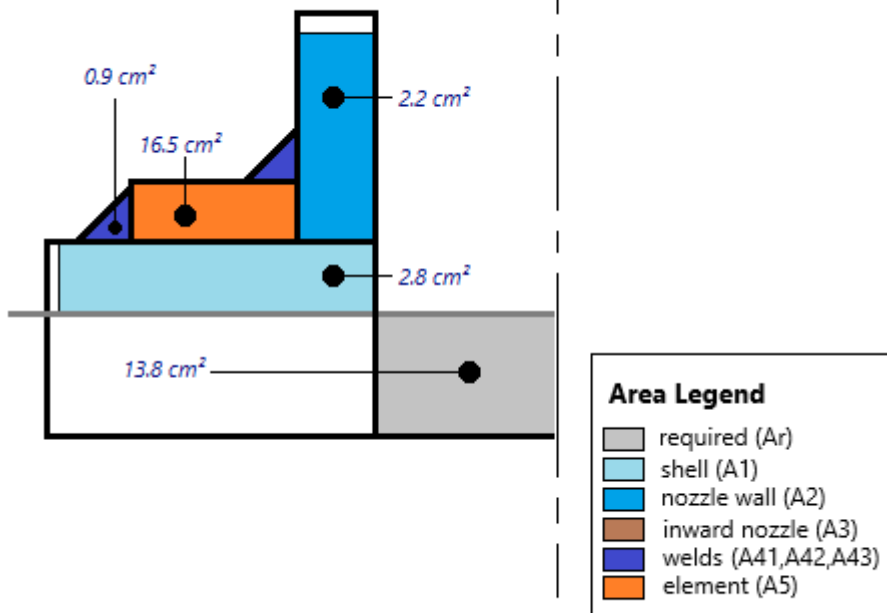


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.755	NA	NA
Area in Shell	A1	2.772	NA	NA
Area in Nozzle Wall	A2	2.189	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	16.450	NA	NA
TOTAL AREA AVAILABLE	Atot	22.267	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 234.4298 12.0000 mm
 Based on given Pad Diameter: 318.2750 5.7910 mm
 Based on Shell or Nozzle Thickness: 240.6228 10.9728 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152.6794 \cdot 9.0093 \cdot 1.0 + 2 \cdot 7.7978 \cdot 9.0093 \cdot 1.0 \cdot (1 - 1.0))$
 $= 13.755 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.679(1.0 \cdot 10.825 - 1.0 \cdot 9.009) - 2 \cdot 7.798$
 $(1.0 \cdot 10.825 - 1.0 \cdot 9.0093) \cdot (1 - 1.0)$
 $= 2.772 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 27.06) \cdot (7.8 - 3.07) \cdot 0.8551$



$$= 2.189 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 0.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (305.3588 - 168.275) \cdot 12.0 \cdot 1.0$
 $= 16.450 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 6.2420 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.6750 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 12.1843 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 12.1843 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.6750 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 9.3950 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.395, \max(12.1843, 4.675)]$
 $= 9.3950 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(6.242, 9.395)$
 $= 9.3950 \text{ mm}$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 209.7,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2798.3 kgf/cm ²	Passed
Occasional	: 219.8,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 46.4,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.601$, $t_r = 3.067$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.477$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.601$, $t_r = 3.067$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.477$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.601$, $t_r = 3.067$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.477$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C



Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio $= tr * E^* / (t_g - c) = 0.832$, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $t_g = 9.601$, $tr = 3.067$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio $= tr * E^* / (t_g - c) = 0.477$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	: -43 °C
Gov. MDMT of the Reinforcement Pad	: -32 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $47.20 / 104.01 = 0.454$

Weld Size Calculations, Description: SH OUT NOZ-1

Intermediate Calc. for nozzle/shell Welds	T_{min}	7.7978	mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	10.8250	mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4585 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$5.4125 = 0.5 * T_{minPad}$	$7.0700 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (A_r - A_1) * S_v$
 $= (13.7553 - 2.7722) * 1407$
 $= 15454.70$ kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 * fr_2) * S_v$
 $= (2.1895 + 16.4501 + 0.8551 - 0.0 * 0.86) * 1407$
 $= 27431.48$ kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) * D_{lo} * W_o * 0.49 * S_{nw}$
 $= (3.1416/2.0) * 168.275 * 10.0 * 0.49 * 1203$
 $= 15584.$ kgf

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) * D_P * W_P * 0.49 * S_{EW}$
 $= (3.1416/2.0) * 318.275 * 10.0 * 0.49 * 1407$
 $= 34471.$ kgf

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) * D_{lo} * W_{gpn} * 0.74 * S_{eg}$
 $= (3.1416/2) * 168.275 * 12.0 * 0.74 * 1407$



= 33028. kgf

Shear, Nozzle Groove Weld [Sngw]:

$$= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng$$

$$= (3.14/2) \cdot (152.679+10.973) \cdot (10.973-3.175) \cdot 0.6 \cdot 1407$$

$$= 16924. kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNGW) = (34471 + 16924) = 51395 kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 51394 kgf, must exceed W = 15454 kgf or W1 = 27431 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 56 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 13.9901 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.9901 mm

Input Echo, WRC297 Item 1, Description: SH OUT NOZ-1 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	520.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	10.9728	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	318.275	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	47.20	kgf/cm ²
Radial Load	P	-407.91	kgf
Circumferential Shear	Vc	611.86	kgf
Longitudinal Shear	Vl	611.86	kgf
Circumferential Moment	Mc	0.28	kgf-m.
Longitudinal Moment	Ml	0.37	kgf-m.
Torsional Moment	Mt	0.46	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{\text{Rmean}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{274.587(14.0 - 3.175)}$$

$$= 348.190 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	549.175 mm
Nozzle Outside Diameter	(d) =	168.275 mm
Vessel Thickness used	(T) =	22.825 mm
Nozzle Thickness used	(t) =	7.798 mm
T / t	=	2.927
d / t	=	21.580

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.503$$

Nr/P	=	0.082
Mr/P	=	0.084
M0/P	=	0.082
N0/P	=	0.200
MrD/Mc	=	0.170
NrDL/Mc	=	0.129
M0d/Mc	=	0.183
N0DL/Mc	=	0.148
MrD/M1	=	0.126
NrDL/M1	=	0.097
M0D/M1	=	0.137
N0DL/M1	=	0.285

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	38	-38	38	-38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	521	568	521	568
Outplane Stress Summary	575	546	575	545

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	260	260	260	260
Inplane Stress Summary	306	228	306	227

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10



Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	10	10	-10	-10
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	575	546	576	546

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	38	-38	38	-38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	260	260	260	260
Outplane Stress Summary	314	238	314	237

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	521	568	521	568
Inplane Stress Summary	566	536	567	535

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-10	-10	10	10

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	567	536	567	535

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	527	584	527	584
Hoop Stress Summary	543	599	543	599

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	171	-171	171	-171
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	263	263	263	263
Axial Stress Summary	443	104	446	101

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	29	29	-29	-29

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	551	601	551	601

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	527	584	527	584
Hoop Stress Summary	543	599	543	599

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	171	-171	171	-171
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	263	263	263	263
Axial Stress Summary	443	104	446	101

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------



Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-29	-29	29	29

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	551	601	551	601

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(268.587(14.0 - 3.175))}$$

$$= 346.214 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	537.175 mm
Nozzle Outside Diameter	(d) =	318.275 mm
Vessel Thickness used	(T) =	10.825 mm
Nozzle Thickness used	(t) =	82.798 mm
T / t	=	0.131
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.174$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.080
M0/P	=	0.024
N0/P	=	0.039
MrD/Mc	=	0.251
NrDL/Mc	=	0.098
M0d/Mc	=	0.075
N0DL/Mc	=	0.067
MrD/Ml	=	0.061
NrDL/Ml	=	0.053
M0D/Ml	=	0.019
N0DL/Ml	=	0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1124	1171	1124	1171
Outplane Stress Summary	1187	1136	1187	1135

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	166	-166	166	-166
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	562	562	562	562
Inplane Stress Summary	745	412	746	412

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	11	11	-11	-11

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1187	1136	1187	1136

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	562	562	562	562
Outplane Stress Summary	624	526	625	526

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	166	-166	166	-166
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1124	1171	1124	1171
Inplane Stress Summary	1307	1023	1309	1021

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-11	-11	11	11
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-11	-11	11	11

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 1307 1023 1309 1021

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	521	568	521	568	521	568	521	568
Circ. Pl (SUS)	15	15	15	15	6	6	6	6
Circ. Q (SUS)	38	-38	38	-38	39	-39	39	-39
Long. Pm (SUS)	260	260	260	260	260	260	260	260
Long. Pl (SUS)	6	6	6	6	15	15	15	15
Long. Q (SUS)	39	-39	39	-39	38	-38	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	521.0	568.0	521.0	568.0	521.0	568.0	521.0	568.0
Pm+Pl (SUS)	536.4	583.3	536.4	583.3	527.4	574.3	527.4	574.3
Pm+Pl+Q (Total)	574.4	545.3	574.4	545.3	566.4	535.3	566.4	535.3

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	568.00	1407.13	Passed
Pm+Pl (SUS)	583.32	2110.70	Passed
Pm+Pl+Q (TOTAL)	574.37	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1124	1171	1124	1171	1124	1171	1124	1171
Circ. Pl (SUS)	13	13	13	13	17	17	17	17
Circ. Q (SUS)	49	-49	49	-49	165	-165	167	-167
Long. Pm (SUS)	562	562	562	562	562	562	562	562
Long. Pl (SUS)	17	17	17	17	13	13	13	13
Long. Q (SUS)	166	-166	166	-166	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1124	1171	1124	1171	1124	1171	1124	1171
Pm+Pl (SUS)	1137	1184	1137	1184	1141	1188	1141	1188
Pm+Pl+Q (Total)	1186	1135	1186	1135	1306	1023	1308	1021

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
---------------------	-----------	----------------	--------



Pm (SUS)	1171.00	1407.13	Passed
Pm+Pl (SUS)	1188.20	2110.70	Passed
Pm+Pl+Q (TOTAL)	1308.18	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	527	584	527	584	527	584	527	584
Circ. Pl (SUS)	15	15	15	15	15	15	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	263	263	263	263	263	263	263	263
Long. Pl (SUS)	10	10	10	10	10	10	10	10
Long. Q (SUS)	170	-170	172	-172	170	-170	172	-172
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	527.0	584.0	527.0	584.0	527.0	584.0	527.0	584.0
Pm+Pl (SUS)	545.1	601.6	545.1	601.6	545.1	601.6	545.1	601.6
Pm+Pl+Q (Total)	549.9	600.7	550.0	600.7	549.9	600.7	550.0	600.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	584.00	1203.20	Passed
Pm+Pl (SUS)	601.56	1804.80	Passed
Pm+Pl+Q (TOTAL)	600.69	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: SH OUT NOZ-2

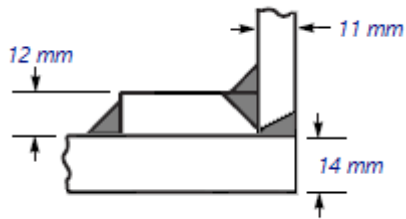
From: 50

Pressure for Reinforcement Calculations	P	47.201	kgf/cm ²
Temperature for Internal Pressure	Temp	88	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	1407.13	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1407.13	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	520.00	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		306.12	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1203.20	kgf/cm ²
Allowable Stress At Ambient	Sna	1203.20	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.9728	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	1407.13	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1407.13	kgf/cm ²
Diameter of Pad along vessel surface	Dp	318.2750	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	10.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		75.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: SH OUT NOZ-2

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.761 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $47.2 \cdot 263.175 / (1407 \cdot 1.0 - 0.6 \cdot 47.2)$
= 9.0093 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $47.2 \cdot 76.34 / (1203 \cdot 1.0 - 0.6 \cdot 47.2)$
= 3.0670 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588 mm
Parallel to Vessel Wall, opening length	d	152.6794 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.0625 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 1203.2 / 1407.1)$
= 0.855

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 1407.1 / 1407.1)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.855, 1.0)$
= 0.855

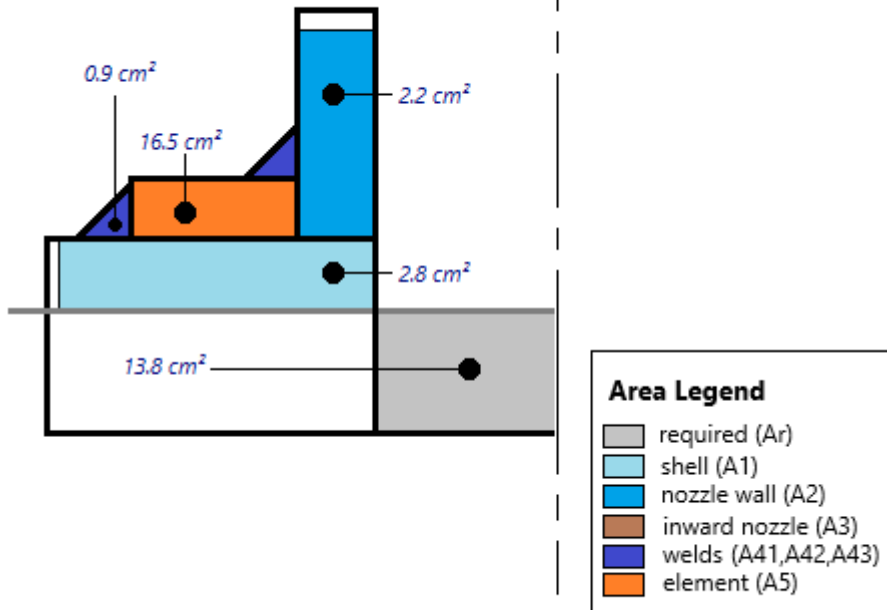


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	13.755	NA	NA
Area in Shell	A1	2.772	NA	NA
Area in Nozzle Wall	A2	2.189	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.855	NA	NA
Area in Element	A5	16.450	NA	NA
TOTAL AREA AVAILABLE	Atot	22.267	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 234.4298 12.0000 mm
 Based on given Pad Diameter: 318.2750 5.7910 mm
 Based on Shell or Nozzle Thickness: 240.6228 10.9728 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (152.6794 \cdot 9.0093 \cdot 1.0 + 2 \cdot 7.7978 \cdot 9.0093 \cdot 1.0 \cdot (1 - 1.0))$
 $= 13.755 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 152.679(1.0 \cdot 10.825 - 1.0 \cdot 9.009) - 2 \cdot 7.798$
 $(1.0 \cdot 10.825 - 1.0 \cdot 9.0093) \cdot (1 - 1.0)$
 $= 2.772 \text{ cm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 27.06) \cdot (7.8 - 3.07) \cdot 0.8551$



$$= 2.189 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:
 $= W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 $= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$
 $= 0.855 \text{ cm}^2$

Area Available in Element [A5]:
 $= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 $= (305.3588 - 168.275) \cdot 12.0 \cdot 1.0$
 $= 16.450 \text{ cm}^2$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 6.2420 \text{ mm}$
Wall Thickness per UG16(b),	$tr16b = 4.6750 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 12.1843 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr16b) = 12.1843 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr16b) = 4.6750 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 9.3950 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[9.395, \max(12.1843, 4.675)]$
 $= 9.3950 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(t_a, t_b)$
 $= \max(6.242, 9.395)$
 $= 9.3950 \text{ mm}$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 10.973 = 9.601 \text{ mm} \rightarrow \text{OK}$$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 209.7,	Allowable	: 1203.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 2798.3 kgf/cm ²	Passed
Occasional	: 219.8,	Allowable	: 1600.3 kgf/cm ²	Passed
Shear	: 46.4,	Allowable	: 842.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 9.601$, $t_r = 3.067$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.477$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 9.601$, $t_r = 3.067$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.477$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 9.601$, $t_r = 3.067$, $c = 3.175 \text{ mm}$, $E^* = 1.0$
Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.477$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, c = 3.175 mm, $E^* = 1.0$
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.832$, Temp. Reduction = 9 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 3.067, c = 3.175 mm, $E^* = 1.0$
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.477$, Temp. Reduction = 36 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-43 °C
Gov. MDMT of the Reinforcement Pad	:	-32 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-32 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $47.20/104.01 = 0.454$

Weld Size Calculations, Description: SH OUT NOZ-2

Intermediate Calc. for nozzle/shell Welds	Tmin	7.7978 mm
Intermediate Calc. for pad/shell Welds	TminPad	10.8250 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4585 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$5.4125 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= (A_r - A_1) \cdot S_v$
 $= (13.7553 - 2.7722) \cdot 1407$
 $= 15454.70 \text{ kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can/.707)^2 \cdot fr_2) \cdot S_v$
 $= (2.1895 + 16.4501 + 0.8551 - 0.0 \cdot 0.86) \cdot 1407$
 $= 27431.48 \text{ kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 1203$
 $= 15584. \text{ kgf}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 318.275 \cdot 10.0 \cdot 0.49 \cdot 1407$
 $= 34471. \text{ kgf}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 168.275 \cdot 12.0 \cdot 0.74 \cdot 1407$



= 33028. kgf

Shear, Nozzle Groove Weld [Sngw]:

$$= (\pi/2) \cdot (Dlr+Wgnva) \cdot (Wgnva-Can) \cdot 0.60 \cdot Sng$$

$$= (3.14/2) \cdot (152.679+10.973) \cdot (10.973-3.175) \cdot 0.6 \cdot 1407$$

$$= 16924. kgf$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNGW) = (34471 + 16924) = 51395 kgf$$

Summary of Failure Path Calculations:

Path 1-1 = 51394 kgf, must exceed W = 15454 kgf or W1 = 27431 kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 56 kgf/cm²

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 13.9901 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.9901 mm

Input Echo, WRC297 Item 1, Description: SH OUT NOZ-2 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	520.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1407.13	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1407.13	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	146.329	mm
Nozzle Thickness	10.9728	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 1203.20	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1203.20	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	318.275	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.

Design Internal Pressure	Dp	47.20	kgf/cm ²
Radial Load	P	-407.91	kgf
Circumferential Shear	Vc	611.86	kgf
Longitudinal Shear	Vl	611.86	kgf
Circumferential Moment	Mc	0.28	kgf-m.
Longitudinal Moment	Ml	0.37	kgf-m.
Torsional Moment	Mt	0.46	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No



Local Loads applied at end of Nozzle/Attachment

No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{274.587(14.0 - 3.175)}$
 $= 348.190 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 549.175 mm
 Nozzle Outside Diameter (d) = 168.275 mm
 Vessel Thickness used (T) = 22.825 mm
 Nozzle Thickness used (t) = 7.798 mm
 $T / t = 2.927$
 $d / t = 21.580$

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.503$$

Nr/P = 0.082
 Mr/P = 0.084
 M0/P = 0.082
 N0/P = 0.200
 MrD/Mc = 0.170
 NrDL/Mc = 0.129
 M0d/Mc = 0.183
 N0DL/Mc = 0.148
 MrD/M1 = 0.126
 NrDL/M1 = 0.097
 M0D/M1 = 0.137
 N0DL/M1 = 0.285

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	38	-38	38	-38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	521	568	521	568
Outplane Stress Summary	575	546	575	545

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	260	260	260	260
Inplane Stress Summary	306	228	306	227

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	10	10	-10	-10



Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	10	10	-10	-10
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	575	546	576	546

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	38	-38	38	-38
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	260	260	260	260
Outplane Stress Summary	314	238	314	237

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	39	-39	39	-39
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	521	568	521	568
Inplane Stress Summary	566	536	567	535

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-10	-10	10	10
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-10	-10	10	10

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	567	536	567	535

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0



Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	527	584	527	584
Hoop Stress Summary	543	599	543	599

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	171	-171	171	-171
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	263	263	263	263
Axial Stress Summary	443	104	446	101

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	29	29	-29	-29
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	29	29	-29	-29

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	551	601	551	601

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	527	584	527	584
Hoop Stress Summary	543	599	543	599

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	10	10	10	10
Axial Bending (P)	171	-171	171	-171
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-1	1	1	-1
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	263	263	263	263
Axial Stress Summary	443	104	446	101

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-29	-29	29	29
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-29	-29	29	29

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	551	601	551	601

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(268.587(14.0 - 3.175))}$
 $= 346.214 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	537.175 mm
Nozzle Outside Diameter	(d) =	318.275 mm
Vessel Thickness used	(T) =	10.825 mm
Nozzle Thickness used	(t) =	82.798 mm
T / t	=	0.131
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.174$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.049
Mr/P	=	0.080
M0/P	=	0.024
N0/P	=	0.039
MrD/Mc	=	0.251
NrDL/Mc	=	0.098
M0d/Mc	=	0.075
N0DL/Mc	=	0.067
MrD/Ml	=	0.061
NrDL/Ml	=	0.053
M0D/Ml	=	0.019
N0DL/Ml	=	0.041

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1124	1171	1124	1171
Outplane Stress Summary	1187	1136	1187	1135

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	166	-166	166	-166
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	562	562	562	562
Inplane Stress Summary	745	412	746	412

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	11	11	-11	-11
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	11	11	-11	-11

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE				
	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	1187	1136	1187	1136

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	49	-49	49	-49
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	562	562	562	562
Outplane Stress Summary	624	526	625	526

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	17	17	17	17
Inplane Bending (P)	166	-166	166	-166
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1124	1171	1124	1171
Inplane Stress Summary	1307	1023	1309	1021

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-11	-11	11	11
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-11	-11	11	11

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside



Two * Max Shear Stress 1307 1023 1309 1021

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	521	568	521	568	521	568	521	568
Circ. Pl (SUS)	15	15	15	15	6	6	6	6
Circ. Q (SUS)	38	-38	38	-38	39	-39	39	-39
Long. Pm (SUS)	260	260	260	260	260	260	260	260
Long. Pl (SUS)	6	6	6	6	15	15	15	15
Long. Q (SUS)	39	-39	39	-39	38	-38	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	10	10	-10	-10	-10	-10	10	10
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	521.0	568.0	521.0	568.0	521.0	568.0	521.0	568.0
Pm+Pl (SUS)	536.4	583.3	536.4	583.3	527.4	574.3	527.4	574.3
Pm+Pl+Q (Total)	574.4	545.3	574.4	545.3	566.4	535.3	566.4	535.3

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	568.00	1407.13	Passed
Pm+Pl (SUS)	583.32	2110.70	Passed
Pm+Pl+Q (TOTAL)	574.37	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1124	1171	1124	1171	1124	1171	1124	1171
Circ. Pl (SUS)	13	13	13	13	17	17	17	17
Circ. Q (SUS)	49	-49	49	-49	165	-165	167	-167
Long. Pm (SUS)	562	562	562	562	562	562	562	562
Long. Pl (SUS)	17	17	17	17	13	13	13	13
Long. Q (SUS)	166	-166	166	-166	49	-49	49	-49
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1124	1171	1124	1171	1124	1171	1124	1171
Pm+Pl (SUS)	1137	1184	1137	1184	1141	1188	1141	1188
Pm+Pl+Q (Total)	1186	1135	1186	1135	1306	1023	1308	1021

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	1171.00	1407.13	Passed
Pm+Pl (SUS)	1188.20	2110.70	Passed
Pm+Pl+Q (TOTAL)	1308.18	4221.40	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	527	584	527	584	527	584	527	584
Circ. Pl (SUS)	15	15	15	15	15	15	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	263	263	263	263	263	263	263	263
Long. Pl (SUS)	10	10	10	10	10	10	10	10
Long. Q (SUS)	170	-170	172	-172	170	-170	172	-172
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	29	29	-29	-29	-29	-29	29	29
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	527.0	584.0	527.0	584.0	527.0	584.0	527.0	584.0
Pm+Pl (SUS)	545.1	601.6	545.1	601.6	545.1	601.6	545.1	601.6
Pm+Pl+Q (Total)	549.9	600.7	550.0	600.7	549.9	600.7	550.0	600.7

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	584.00	1203.20	Passed
Pm+Pl (SUS)	601.56	1804.80	Passed
Pm+Pl+Q (TOTAL)	600.69	3609.61	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description: U-TUBESHEET

Shell Data:

Main Shell Description: SHELL

Shell Maximum Design Pressure	Psd,max	47.20	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	47.20	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	42.84	kgf/cm ²
Shell Thickness	ts	14.0000	mm
Shell Internal Corrosion Allowance	cas	3.1750	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	520.000	mm
Shell Temperature for Internal Pressure	Ts	87.78	°C
Shell Material		SA-516 70	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1407.13	kgf/cm ²
Shell Allowable Stress at Ambient		1407.13	kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	36.25	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	36.25	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	4.50	kgf/cm ²
Channel Thickness	tc	14.0000	mm
Channel Internal Corrosion Allowance	cac	3.1750	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	520.000	mm
Channel Design Temperature	TEMPC	65.56	°C
Channel Flange Matl		SA-516 70	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	1407.13	kgf/cm ²
Channel Allowable Stress at Ambient		1407.13	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	228	
Tube Wall Thickness	et	2.1100	mm
Tube Outside Diameter	D	19.0500	mm
Total Straight Tube Length	Lt	3042.00	mm
Straight Tube Length (bet. inner tubsht faces) L		2952.00	mm
Design Temperature of the Tubes		87.78	°C
Tube Material		SA-210 A-1	
Tube Material UNS Number		K02707	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		1203.20	kgf/cm ²
Tube Allowable Stress At Ambient		1203.20	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2400.75	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	23.8120	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.1100	mm
Groove Weld Leg	ag	2.1100	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		UW-20.3	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	243.630	mm
Largest Center-to-Center Tube Distance	Ul	33.0500	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	6.0000	mm

Tubesheet Data:



Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 87.78 °C
 Tubesheet Material SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K02700
 Tubesheet Allowable Stress at Temperature S 1407.13 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1407.13 kgf/cm²
 Thickness of Tubesheet h 90.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 3.1750 mm
 Tubesheet Corr. Allowance (Channel side) Catc 3.1750 mm
 Tubesheet Outside Diameter A 595.000 mm

Dimension G for the Channel Side Gc 571.582 mm
 Area of the Untubed Lanes AL 167.3 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Both
 Flange Outside Diameter A 698.000 mm
 Flange Inside Diameter B 520.000 mm
 Flange Face Outside Diameter Fod 598.000 mm
 Flange Face Inside Diameter Fid 520.000 mm
 Gasket Outside Diameter Go 592.000 mm
 Gasket Inside Diameter Gi 526.000 mm
 Small end Hub thk. g0 14.0000 mm
 Large end Hub thk. g1 20.0000 mm
 Gasket Factor, m 3.75
 Gasket Design Seating Stress y 534.33 kgf/cm²
 Flange Facing Sketch Code Sketch 1a
 Gasket Thickness tg 3.1750 mm
 Full face Gasket Flange Option Program Selects
 Length of Partition Gasket lp 526.000 mm
 Width of Partition Gasket wp 10.0000 mm
 Partition Gasket Factor, mPart 0.0000
 Partition Gasket Design Seating Stress yPart 0.00 kgf/cm²
 Partition Gasket Facing Sketch Code Sketch 1a
 Partition Gasket Column for Gasket Seating Code Column II

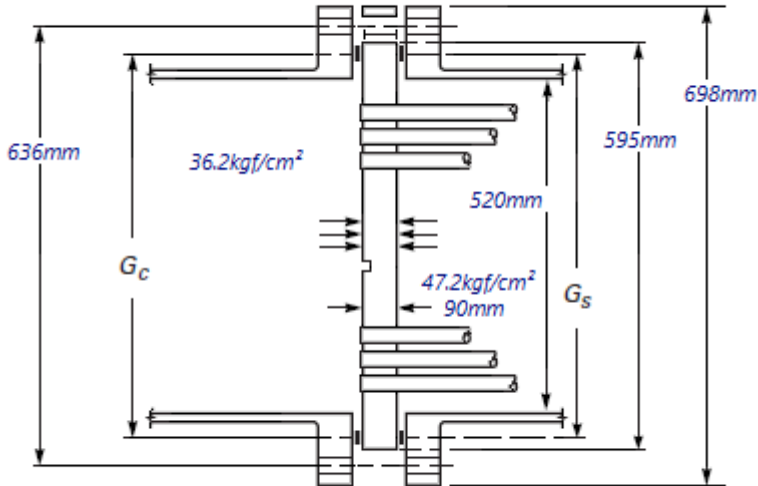
Bolting Information:

Diameter of Bolt Circle C 636.000 mm
 Nominal Bolt Diameter dB 27.0000 mm
 Thread Series TEMA Metric Thread
 Number of Bolts n 28
 Bolt Material SA-193 B7
 Bolt Allowable Stress At Temperature Sb 1753.82 kgf/cm²
 Bolt Allowable Stress At Ambient Sa 1753.82 kgf/cm²
 Weld between Flange and Shell/Channel 0.0000 mm

Alternate Flange Operating Bolt Load, Wm1 186130.50 kgf
 Alternate Flange Seating Bolt Load, Wm2 98169.09 kgf
 Alternate Flange Design Bolt Load, W 203229.78 kgf

Tubesheet Integral with None
 Tubesheet Extended as Flange No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	87.8 °C	0.20281E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	65.6 °C	0.20411E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	87.8 °C	0.20281E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	87.8 °C	0.20281E+07 kgf/cm²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (36.25 \cdot (19.05/2 - 0)) / (1203.2 \cdot 1.0 + 0.4 \cdot 36.25)$
 $= 0.2836 + 0.0000 = 0.2836 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-2 at 87.78 °C
 Elastic Modulus for Material 2038903.00 kgf/cm²

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	3195.63	9.03	50.0000	0.0134948	1251.46
EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 196.1288 \text{ kgf/cm}^2$						

Results for Req'd Thickness for Ext. Pressure (Tca):



TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6904	19.05	3195.63	27.59	50.0000	0.0014447	976.93

EMAWP = (4*B)/(3*(D/T)) = (4 *976.9268)/(3 *27.5935) = 47.2057 kgf/cm²

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6904 mm
Allowable Internal Pressure at Corroded thickness	292.45 kgf/cm ²
Required Internal Design Pressure	36.25 kgf/cm ²
Allowable External Pressure at Corroded thickness	196.13 kgf/cm ²
Required External Design Pressure	47.20 kgf/cm ²
Required Thickness due to Shell Side pressure	0.6904 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (Goc-Gic) / 2	33.000 mm
Basic Gasket Width,	b0 = N / 2	16.500 mm
Effective Gasket Width,	b = sqrt(b0) * 2.5	10.232 mm
Gasket Reaction Diameter,	G = Go-2.0*b	571.536 mm

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
= 2 • ro + dt
= 2 • 243.63 + 19.05
= 506.310 mm

Basic ligament efficiency for shear [μ]:
= (p - dt) / p
= (23.812 - 19.05)/23.812
= 0.200

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:
= Gs / Do (Configurations d, e, f)
= 571.5355/506.31
= 1.1288

Ratio [pc]:
= Gc / Do (Configurations d)
= 571.5824/506.31
= 1.1289

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:
= ltx / h
= 0/90.0
= 0.000 (must be 0 ≤ p ≤ 1)

Effective tube hole Diameter [d*]:
= max(dt - 2tt(EtT/E) • (StT/S) • (p); dt - 2tt)
= max(19.05 - 2 • 2.11 (2028090/2028090) •
(1203/1407) • (0); 19.05 - 2 • 2.11)
= 19.0500 mm

Effective tube pitch [p*]:
= p / √(1 - 4 • min(A1; 4 • Do • p)/(π • Do²))
= 23.812/√(1 - 4 • min(167.33 ; 4 • 506.31 • 23.812)/(π • 506.31²))



= 24.8678 mm

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (24.8678 - 19.05) / 24.8678 \\ &= 0.2339 \end{aligned}$$

$h/p = 3.779607$; $\mu^* = 0.233949$; $E^* = 398491 \text{ kgf/cm}^2$
 $E^*/E = 0.196486$; $v^* = 0.414066$

Note: As h/p (3.780) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 595.0 / 506.31 \\ &= 1.1752 \end{aligned}$$

Coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (E \cdot \ln(K)) \\ &= (1 - 0.41) / 398491 \cdot (2028090 \cdot \ln(1.18)) \\ &= 0.4813 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s , P_t) [Mts]:

$$\begin{aligned} &= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t] \\ &= 506.31^2 / 16 \cdot [(1.129 - 1) \cdot (1.129^2 + 1) \cdot 47.2 - \\ &\quad (1.129 - 1) \cdot (1.129^2 + 1) \cdot 36.25] \\ &= 51262.7461 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M^*]

$$\begin{aligned} &= Mts + W^* \cdot (G_c - G_s) / (2 \cdot \pi \cdot D_o) \\ &= 51262.7 + 186130.5 \cdot (571.582 - 571.582) / (2 \cdot \pi \cdot 506.31) \\ &= 51262.7461 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [M_p]:

$$\begin{aligned} &= (M^*) - D_o^2 / 32 \cdot F (P_s - P_t) / (1 + F) \\ &= (51262.75) - 506.31^2 / 32 \cdot 0.481 (47.2 - 36.25) / (1 + 0.48) \\ &= 6102.1348 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [M_o]:

$$\begin{aligned} &= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t) \\ &= 6102.13 + 506.31^2 / 64 \cdot (3 + 0.414) \cdot (47.2 - 36.25) \\ &= 155843.2500 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(|6102.135|; |155843.25|) \\ &= 155843.2500 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - h_g')^2) \\ &= 6 \cdot 155843.25 / (0.2339 (90.0 - 6.0)^2) \\ &= 566.4479 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1407.13 \\ &= 2814.27 \text{ kgf/cm}^2 \end{aligned}$$



Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 155964.594 / (0.2339 (43.716 - 6.0)^2) \\ &= 2811.9370 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + \text{Cats} + \text{Catc} \\ &= 43.716 + 0 + 0 \\ &= 43.7160 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(\text{HreqB}; \text{HreqS}) \\ &= \max(43.716 ; 6.1568) \\ &= 43.7160 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |47.2 - 36.25| \\ &= 10.950 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.2 \cdot 90.0 / 506.31 \cdot \min(0.8 \cdot 1407.134 ; 0.533 \cdot 2461.931) \\ &= 160.068 \text{ kgf/cm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot |P_s - P_t| + \text{Cats} + \text{Catc} \\ &= 1 / (4 \cdot 0.2) \cdot (506.31 / 1125.707) \cdot |47.2 - 36.25| + 0 \\ &= 6.1568 \text{ mm} \end{aligned}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (19.05 - 2.11) 2.11 \\ &= 1.123 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.11 (19.05 - 2.11) 1203.202 \\ &= 1351.088 \text{ kgf} \end{aligned}$$

Groove Weld Strength [Fg]:

$$\begin{aligned} &= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t) \\ &= 0.85 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 1203.2 \\ &= 1351.0878 \text{ kgf} \end{aligned}$$

Weld Strength Factor [fw]:

$$\begin{aligned} &= S / S_w \text{ where } S_w = \min(S \text{ and } S_t) \\ &= 1203.202 / 1203.202 \\ &= 1.000 \end{aligned}$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$\begin{aligned} &= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t) \\ &= 0.55 \cdot \pi \cdot 2.11 (19.05 + 0.67 \cdot 2.11) 1203.2 \\ &= 897.6708 \text{ kgf} \end{aligned}$$

Minimum Length of Weld [ar]:

$$\begin{aligned} &= 2 \sqrt{ ((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d) - 0.75 \cdot d_o } \\ &= 2 \sqrt{ ((0.75 \cdot 19.05)^2 + 1.07 \cdot 2.11 \cdot (19.05 - 2.11) 1.0 \cdot 1.0) - } \end{aligned}$$



$0.75 \cdot 19.05$)
 = 2.562 mm

Minimum Required Fillet Weld Leg [afr]: 1.2810 mm
 Minimum Required Groove Weld Leg [agr]: 1.2810 mm

Tube-Tubesheet Jt allowable, 1351.09 is >= tube strength 1351.09 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	566.4 <=	2814.3 kgf/cm ²	Ok
Tubesheet shear stress	77.0 <=	1125.7 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	43.716	90.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.281	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.281	2.110	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	74.693 4.235	1877 2814	255 1126	Ps+Pt D1	Ok
D2uc	84.297 5.514	2443 2814	332 1126	Ps+Pt D2	Ok
D3uc	43.716 1.279	566 2814	77 1126	Ps+Pt D3	Ok
D1c	77.787 4.237	2027 2814	274 1126	Ps+Pt-c D1	Ok
D2c	87.444 5.517	2639 2814	357 1126	Ps+Pt-c D2	Ok
D3c	46.886 1.280	612 2814	83 1126	Ps+Pt-c D3	Ok
Max:	87.4435 5.517 mm	0.938	0.317 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	87.444	90.000	Ok
Tube thickness :	0.690	2.110	Ok



Tube-tubesheet fillet weld leg :	1.281	2.110	Ok
Tube-tubesheet groove weld leg :	1.281	2.110	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	50.328	1.000	50.337	1.000
Tubesheet Shear Stress	148.773	1.000	148.773	1.000
Tube Pressure Stress	292.449	1.000	196.128	1.000
Minimum MAWP	50.328		50.337	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	54.360	1.000	54.360	1.000
Tubesheet Shear Stress	160.067	1.000	160.067	1.000
Tube Pressure Stress	292.449	1.000	196.128	1.000
Minimum MAPnc	54.360		54.360	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):
 = tubesheet thickness/4
 = 90.0/4
 = 22.500 mm

Thickness Ratio = 0.868, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -11 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(36.25/54.36, 47.2/54.36)
 = 0.868

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PV Elite[®] Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: